

Analysis of Syntax Structure's Impact on Fact Retrieval from Pre-trained Language Models

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Declaration

I hereby declare that I have written and created this Master Thesis on my own and that I did not use any other than the stated sources and aids.

Düsseldorf, 10th of March 2023

A handwritten signature in black ink, appearing to read 'Tim Tressel', written in a cursive style.

Tim Tressel

Abstract

Large Pre-trained language models (PLMs) like BERT and its successors have caused an explosion of innovation in the field of natural language processing because their special architecture has caused the reaching of unprecedented heights in performance on many important tasks. Inspired by Petroni et al. (2019), the ability of PLMs to function as knowledge bases is widely researched. Especially, the engineering and optimization of prompts for fact retrieval is a topic of large research interest. In this investigation, we focus on the relationship between the syntactical structure of prompts and fact retrieval performance. We construct syntactically different but semantically equivalent prompts in order to assess the relationship between syntax and fact retrieval performance of PLMs. We investigate two syntactical frameworks, the sentence typology and the active, passive and nominalized verb morphology to construct semantically equal but syntactically different prompts. We conclude that BERT-based PLMs generally do not generate the same output to syntactically different but semantically similar sentences. All investigated models have clear syntactical favorites. Furthermore, we show that neither superficial properties of the prompts nor syntax dependency graphs can explain the performance variance.

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1 Introduction

The invention of the transformer based models and other subsequent attention models like BERT and its successors have caused an explosion of research and innovation in the field of natural language processing because their special architecture has caused the reaching of new heights in performance on many important tasks, such as text generation, language understanding, translation and many others. One interesting avenue of research that has since been explored is the use of language model as knowledge bases. One such model called ChatGPT has recently gained a lot of main stream attention and will surely amplify interest in this topic further. The construction of large knowledge bases has been an endeavor undertaken by computer scientists in order to make either the world's knowledge or specific subsets of it automatically accessible for computer programs. However, maintaining their accuracy and completeness remains a great challenge. Additionally, querying knowledge graph based knowledge bases rely on hard to learn query languages. If pre-trained neural network based language models, hence called PLMs, could perform equally well or even better than current knowledge bases, this development would solve both of these issues. A PLM that was trained on Wikipedia can accurately predict the next words in a sentence, could it therefore give a correct answer to a question? If so, does it depend on how the question is posed, and can a PLM be trained in a manner that resolves this dependency? This work aims to narrow this broad question down to the aspect of syntactical structure. The research question this work tries to answer is as follows. Can PLMs generate the same output to syntactically different but semantically similar sentences? The experiments undertaken in this investigation lead to the conclusion that PLMs generally do not generate the same output to syntactically different but semantically similar sentences, and we attempt to demonstrate when they do and when they do not as well as why these discrepancies exist. Moreover, we will find out that all PLMs investigated in this work clearly favor some syntactical structure over others, and we furthermore demonstrate that these differences are not explained by superficial properties of sentences, such as prominent positions of words, the amount of keywords in sentences or the length of sentences. The differences are caused by the differences in syntax that influence the way a model processes and interprets language. In order to reach this objective, we will introduce the relevant theory behind modern PLMs, discuss the relevant research about using language models as knowledge bases and introduce the linguistic fundamentals required for our analysis. Furthermore, we will devise and describe the experiments undertaken to investigate the research question and analyze and discuss the results.

2 Language Models

A language model is defined as a model describing the probability of a given sequence of words given a subsequence of these words as input. Often these sequences of words are a sentence or an entire text. Language models are then employed to predict one or more of the following words based on the probabilities of the words in their vocabulary. In this work, we will use the other, more practical and derivative meaning of the term language model as a function that can assign a probability to a next or a masked word in

a sentence. Although there are many possible ways to construct language models neural network based language models have dominated the field for some time now and transformer based attention models have in turn dominated the field since their inception in 2017. Neural language models have been around and popular since the 1980s and until very recently the two main forms were the well known sequence to sequence architectures the Recurrent Neural Network(Rumelhart et al., 1986), short RNN, and the Long short-term memory(Hochreiter and Schmidhuber, 1997), short LSTM. Both of these architectures were trained sequentially, meaning they processed one word at a time and carried a hidden state over to the processing of the next word. These architectures were very successful, but also had significant drawbacks regarding the training time and the quality of the results they produced. Effective language models are the reason for the explosion of innovation and research in the field of natural language processing. Since a language model is as the name suggests a model of language which means it is fundamentally a probability distribution that can assign a specific probability to a sentence, a randomly initialized neural architecture capable of learning a probability distribution is usually trained on a large text corpus.

2.1 Theory of PLMs

In the now famous paper "Attention is all you need" Vaswani et al. (2017) introduced the world to the transformer models of language processing which is considered a major breakthrough and has since its first publication led to many further great developments such as BERT (Devlin et al., 2018), and GPT3 (Brown et al., 2020). Up until the point of this paper the dominant forms of neural language models, which are sequence transduction models, were recurrent or convolutional neural networks that connect an encoder and a decoder through some attention mechanism. As the name of the paper suggests, Vaswani et al. (2017) propose that transformers built exclusively upon attention mechanisms outperform their predecessors while being significantly faster to train due to a more parallel architecture.

2.1.1 Word Embeddings

In order to make computers process words, these must be numerical-ized and numerous transformation strategies have been developed over the years. Various preprocessing methods have been conceived and employed in order to sanitize and standardize words into tokens. Afterward, virtually all methods start with a fixed-sized and enumerated vocabulary and map each token contained in this vocabulary to a fixed index. From this point on, further transformations into vector spaces can follow. The most common methods are One-Hot-Encoded Vectors, the Word-Document-Matrix, Word-Word-Co-Occurrence-Matrix as well as neural network based learned embeddings. All models that are discussed in this work rely on neural network based embeddings, since these embeddings can learn to encode the exact information that is needed for the specific task. PLMs like BERT usually provide custom embeddings tailored specifically to their requirements.

2.1.2 Directional Language Models

Generally, there are two different kinds of language models, unidirectional language models and bidirectional language models. Unidirectional language models are probability distributions that assign probabilities to sequences of words, word combinations, entities etc. represented by tokens. A sequence of tokens of length n , represented by a vector $w = [w_1, w_2, \dots, w_n]$ is assigned a probability $p(w)$ by factoring the individual tokens' conditional probabilities conditioned on the tokens predecessors

$$p(w) = \prod_t^n p(w_t | w_{t-1}, \dots, w_1). \quad (1)$$

Bidirectional language models in contrast use not only a tokens predecessors but also its successors and thus use the entire context available to a single token

$$p(w_t) = p(w_t | w_n, \dots, w_{t+1}, w_{t-1}, \dots, w_1). \quad (2)$$

2.1.3 Transformers

The point of departure for transformer models from their precursors lies in the reliance on recurrence. Recurrence has presented an issue because of the sequence length constraint, inherent to all recurrent approaches, which precludes parallelization. The Transformer in contrast is the first transduction model based exclusively on self-attention to compute representations of input and output without using sequence-aligned recurrence or convolution. In order to explain how the transformer works three aspects of its architecture, see Figure 1, must be explained, the encoder, the decoder and the multi-head self-attention mechanism.

Like many other sequence models, the transformer is also composed of two parts, the encoder and the decoder. The encoder consists of multiple identical layers which are in turn constructed using two separate parts, first a multi-head self-attention mechanism and second a position-wise fully connected feed-forward network. State-of-the-art improvements like normalization and residual connections are also employed. The decoder also consists of multiple identical layers, but each of its layers is constructed from three parts as opposed to just two. The decoder also uses the multi-head self-attention mechanism and a position-wise fully connected feed-forward network, but interleaves them with a third layer performing multi-head attention over the output of the encoder.

The multi-head self-attention mechanism is based on an attention function, which is just a name for a vector-valued function that maps a query and a set of key-value pairs to a sum of values weighted by weights based on the query and keys. Vaswani et al. (2017) use a concept they call *Scaled Dot-Product Attention* because they use the standard Dot-Product Attention, which means they take the Softmax of the dot product of query and keys before performing a pointwise multiplication with the values but scale this dot product with the square root of the dimension of the key vector, as seen in Figure 2. As a batchwise operation multiple outputs can be calculated in parallel using the following equation where Q, K, V are matrices composed of query, key and value vectors

$$(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (3)$$

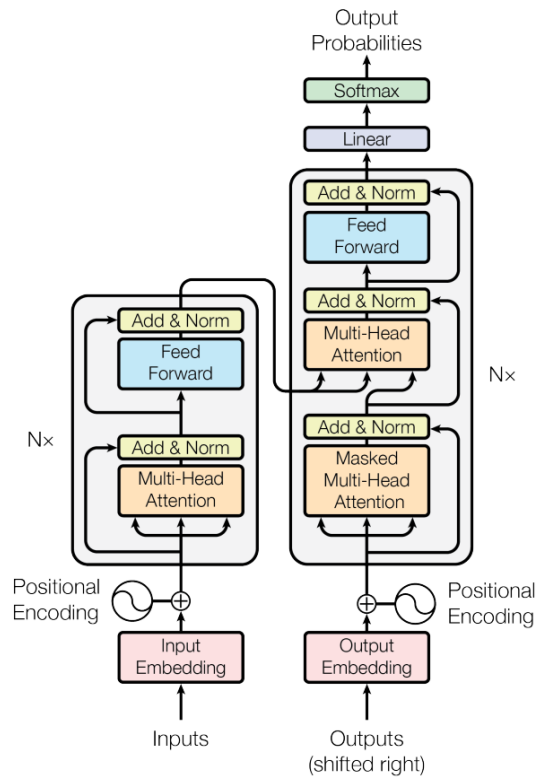


Figure 1: Transformer Architecture (Vaswani et al., 2017)

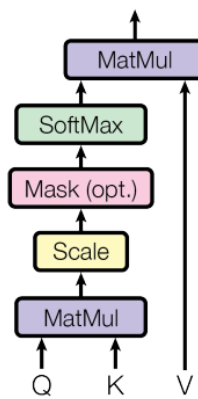


Figure 2: Scaled Dot-Product Attention (Vaswani et al., 2017)

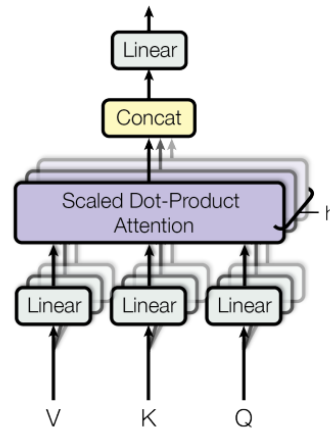


Figure 3: Multi Head Attention (Vaswani et al., 2017)

The multi-head self-attention mechanism now performs this operation not on the queries, keys and values directly but uses several, at least three, different linear projections then performs Scaled Dot-Product Attention on these projections concurrently, concatenates the results and linearly projects them again into a final output, as seen in Figure 3. The results, based on English-to-German and English-to-French translation, presented by Vaswani et al. (2017) for the superiority of using transformers over previous methods both in accuracy and training speed are very compelling. These discoveries are all very impactful in the field of natural language processing because the primary application of sequence modeling are language modeling and machine translation.

2.1.4 BERT

Devlin et al. (2018) introduced the world to BERT, an acronym for *Bidirectional Encoder Representations from Transformers*, which is a transformer model intended for PLMs. The innovation BERT brings to the table is the fact, that it is the first transformer model that is trained in a bidirectional manner, which means it is not strictly trained from left to right or from right to left in an autoregressive manner. In order to achieve this bi-directionality, the authors use a concept they call a masked language model, which masks certain words in a sentence and thus allows the model to direct attention to words positioned before and after the mask in the input sequence. Furthermore, during training BERT does not only predict a token from its vocabulary for the masked token in a sentence, but BERT further must decide whether a randomly sampled sentence from the training text corpus is the next sentence in the text corpus or not.

As stated above BERT is a transformer model and the architecture is very similar to the original transformer introduced by Vaswani et al. (2017) and described above but since the BERT model can process all words in two sentences simultaneously a normal input embedding is not sufficient, thus the words are embedded using a pre-trained token-embedding, a segment embedding to distinguish between the two input sentences and a positional encoding to encode the positions of words within the sentences. As shown in Figure 5 these three embedding vectors of a word are summed up and yield the vector

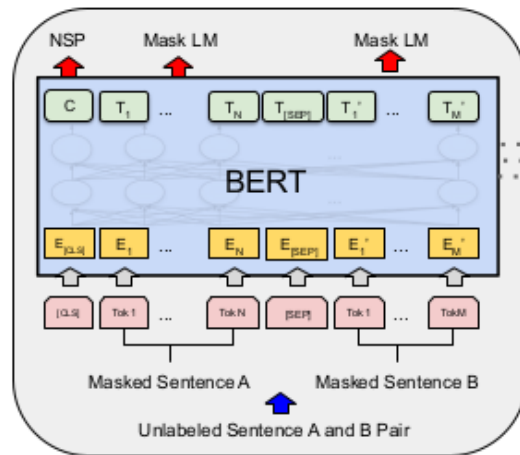


Figure 4: BERT - Pre-training architecture

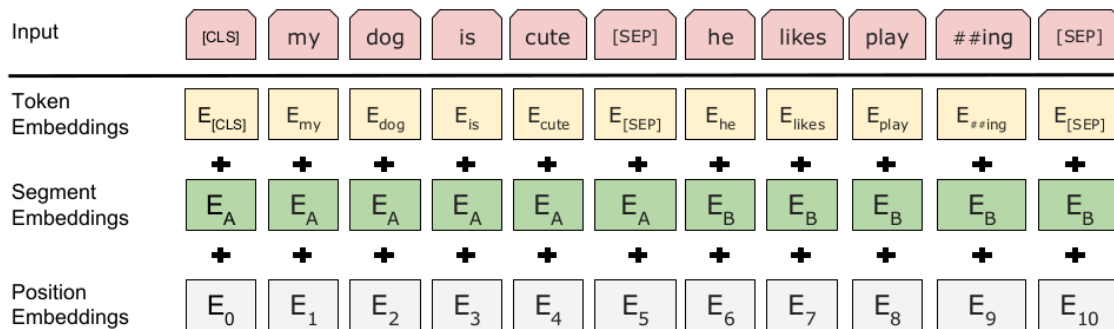


Figure 5: BERT - input embedding

embedding of a single word. These embeddings for two separate sentences are then given as input to the model. Mirroring the input, the model also outputs two sentences as a set of embedding vectors and an additional percentage. For each input token, there is a single vector contained in the set of embedding vectors. The percentage value in the output is supposed to signify the model's confidence that the second sentence directly follows the first. Finally, fine-tuning BERT for a specific task just requires one to choose an appropriate architecture for the downstream output and training the new parameters, for example predicting the correct token for the mask or translating a sentence into a different language.

BERT is a pre-trained language model, and pre-training is the process of training a model on unlabeled data in preparation of actual tasks focused on labeled data. Pre-training can be applied in two distinct approaches, one is called *feature based*, the other one is called *fine-tuning*. In the feature based approach, the existing model is extended by integrating the pre-trained architecture to obtain additional features to use in the final layers. The fine-tuning approach uses the opposite strategy and extends the pre-trained architecture by adding a minimal set of task-specific parameters, yielding a new different form of output. The pre-training of BERT consists of two separate phases. The mask language model and the next sentence prediction phases are both performed simultaneously and

both depicted in Figure 4. For the mask language model training, a certain percentage of the input is hidden from the model and mask tokens are used instead. The model is then trained to predict the correct words behind the masks. Simultaneously, the model is trained for next sentence prediction by being fed a second sentence and trying to predict whether it is actually the next sentence in the text corpus, which it is half of the time and otherwise a random sentence from the corpus. There are four different forms of pre-trained BERT models available for the English language. There is BERT-base and a BERT-large which are differentiated by the number of encoder layers, BERT-base has 12 and BERT-large has double the amount with 24. Also, there are cased and uncased versions of BERT, differentiated by their vocabulary size. The cased version differentiate between capitalized and lower cased words, whereas the uncased versions do not. Furthermore, there is a multilingual version of BERT, that can also be used to process text in English. This gives a choice of six different BERT-based models:

1. BERT-base-uncased
2. BERT-base-cased
3. BERT-large-uncased
4. BERT-large-cased
5. BERT-base-multilingual-uncased
6. BERT-base-multilingual-cased

The results presented by Devlin et al. (2018) obtained from the experiments with the various datasets are very compelling because they outperform all previous benchmarks, including Open AI's GPT, by a substantial margin, even with comparatively little training. These discoveries are all very impactful in the field of natural language processing because BERT's bidirectional architecture can be used to improve the performance of a multitude of natural processing tasks.

2.1.5 SenseBERT

Also, attempting to improve BERT, Levine et al. (2019) introduce a new flavor of BERT called SenseBERT by adding a weak form of supervision to the training in order to add further information, "sense", hence the name, to the model by combining words with their WordNet Supersenses. Since any word is only a surrogate that can potentially have numerous different meanings in varying contexts, the sense of the word refers to the actual meaning of a word in the context it is used in. Thus, the authors hope to attain a lexical semantic level language model, without the use of human annotation. SenseBERT is derivative from BERT in the sense that it shares most of BERT's architecture, but with a couple of key differences. First, SenseBERT only uses the mask prediction aspect of the pre-trained BERT and dispenses with the next sentence prediction. Second, it adds the prediction of the WordNet Supersenses. In concrete terms, this means that after the encoding by BERT, the encoded vector embeddings of the input are forwarded to two different linear layers with a Softmax each. The first predicting the masked token and

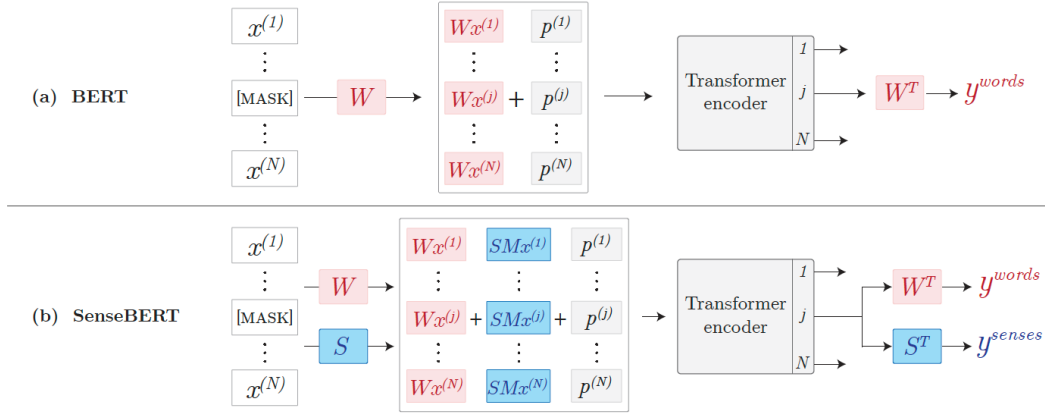


Figure 6: SenseBERT vs. BERT - Architecture Comparison (Levine et al., 2019)

the second predicting the Supersense. This architecture in comparison to vanilla BERT is shown in Figure 7.

In order to construct the labels for the training data, the correct Supersense for a masked word must be found. This label is created as the union of Supersenses coupled with the Sysnsets that were retrieved from WordNet. The training of SenseBERTs adds a new cross-entropy loss term for the Supersenses to BERT’s standard mask prediction cross-entropy loss and a necessary special regularization term to account for the problem of only prediction the broadest possible Supersense. Let $A(w)$ denote the union of Supersenses WordNet assigns for the word w then the following regularization term is used to ensure the prediction of all relevant Supersenses

$$\mathcal{L}_{reg} = - \sum_{s \in A(w)} \frac{1}{|A(w)|} \log p(s|c) \quad (4)$$

and where c stands for the context of word w and p is the probability of one of the Supersenses given by the models second Softmaxed output.

2.1.6 Electra

Clark et al. (2020) introduce Electra, which stands for *Efficiently Learning an Encoder that Classifies Token Replacements Accurately*. It is a further improvement over BERT based on the authors’ insight, that masked language models are essentially de-noiser models where a sentence is noised by masking a single token. Therefore, the model only has the opportunity to learn from a single token, the missing one. As a solution, they propose pre-training a language model by training it to find corrupted tokens that were sampled in training. This means that the model can learn from all tokens, causing greater learning efficiency. The improvement over BERT comes from the fact that it outperforms BERT substantially given the same model size, data, and compute. This greater efficiency makes it possible to train a small version of Electra on a single GPU in only four days.

Electra’s architecture is, like SenseBERT is too, very similar to that of BERT it consists of the encoder side of a transformer-encoder-decoder architecture. Electra is trained as a discriminator in a generator-discriminator architecture, not completely unlike a general adversarial network. A typically very small masked language model is used to replace a couple of sampled masked tokens and the discriminator, which is the actual model that is used for later downstream tasks, is trained to recognize and label all input tokens as original ones or those that have been replaced by the generator.

Let h_G denote the transformer-based encoder function of the generator and e denote the token embeddings function then for an input sequence $x = [x_1, \dots, x_t, \dots, x_n]$ of length n which is masked in position t the probability for a token y to be the originally masked token can be calculated the following way

$$p_G(y|x) = \frac{\exp(e(y)^T h(x)_t)}{\sum_{k=1}^n \exp(e(x_k)^T h(x)_t)}. \quad (5)$$

The generator transforms a sequence of tokens into encoded vectors representations followed by linear and Softmax layers in order to return token probabilities for masked tokens in the input sequence. The tokens with the highest probability replace the masked tokens in the input sequence, and this new corrupted token sequence is then given to the discriminator. This generator is trained as a normal masked language model to accurately predict the masked token in the original input sequence. The discriminator is trained with the objective of differentiating between the tokens that are corrupted and those that are not. This is different in the sense that the discriminator needs to make a binary decision for every token. In order to express the discriminator’s confidence that a specific single token in its input sequence was corrupted as a probability, a sigmoid function is used. Let h_D denote the discriminator’s transformer-based encoder function, ω the weights of the discriminator’s linear layer and t as the index of the position of a token then discriminator will calculate the probability of the corruption of position t token with the following function

$$p_D(x^{corrupt}, t) = \frac{1}{1 + \exp(\omega^T h_D(x_t^{corrupt}))}. \quad (6)$$

In order to train the generator and the discriminator in parallel, two loss functions are necessary. One loss for the generator and one loss for the discriminator. The generator is trained using a maximum likelihood optimization. The training process works the following way. Given a sequence $x = [x_1, \dots, x_n]$ of length n and a number of tokens to be replaces by masks, expressed as the hyperparameter k , we first sample k random indices from a uniform probability distribution from 1 to n , recorded in $m = [m_1, \dots, m_k]$

$$m_i \sim \text{uniform}(1, n) \quad (7)$$

Then we mask the tokens with indices in m

$$x^{masked} = \text{replace}(x, m, [MASK]). \quad (8)$$

Afterwards we apply the generator function to the masked sequence, which can be expressed as sampling k tokens from p_G for index $i \in m$

$$\hat{x}_i \sim p_G(x_i | x^{masked}). \quad (9)$$

With the replacement tokens in $\hat{x}$ we finally arrive at the corrupted sequence

$$x^{corrupt} = \text{replace}(x, m, \hat{x}). \quad (10)$$

With these terms at hand, we define the loss function for the generator

$$\mathcal{L}_G(x, \theta_G) = \mathbb{E} \left(\sum_{i \in m} -\log(p_G(x_i | x^{masked})) \right) \quad (11)$$

and the loss of the discriminator

$$\mathcal{L}_D(x, \theta_D) = \mathbb{E} \left(\sum_{t=1}^n -\mathbb{1}(x_t^{corrupt} = x_t) \log(p_D(x^{corrupt}, t)) - \mathbb{1}(x_t^{corrupt} \neq x_t) \log(1 - p_D(x^{corrupt}, t)) \right). \quad (12)$$

The terms θ_G and θ_D stand for the complete set of learnable parameters for the generator, θ_G , and the discriminator, θ_D . Combining these two losses lets us arrive at the ultimate combined loss for a large text corpus $\mathcal{X}$

$$\mathcal{L}_{ELECTRA} = \min_{\theta_G, \theta_D} \sum_{x \in \mathcal{X}} \mathcal{L}_G(x, \theta_G) + \lambda \mathcal{L}_D(x, \theta_D). \quad (13)$$

In practice, the expected values in the losses are only evaluated with a single sample. Furthermore, the sampling step impedes the backpropagation through generator and discriminator at the same time. After training, the generator is discarded and the now pre-trained discriminator is kept to be fine-tuned on down-stream tasks like translation, question answering or masked token prediction. For all the similarities to the training of a generative adversarial network, there are three essential differences. One is the above-mentioned fact that sampling impedes combined backpropagation, second there is no noise vector as input to the generator and finally the generator is not trained to fool the discriminator. This is exemplified by the fact, that if the generator creates the correct token for the mask, it is treated as if the token was not corrupted. The authors experimented with several approaches to change these properties, but found them to have diminished performance.

2.2 Analysis of PLMs

In order to investigate the impact of syntactical structure on fact retrieval performance of PLMs, we need to define what we mean by performance and introduce the metrics that we want to utilize to measure, compare and understand the results.

2.2.1 Token Distributions

When a PLM of the masked language model variety, like BERT, is queried with a masked prompt, the output is a score vector of the size of the vocabulary size and a Softmax

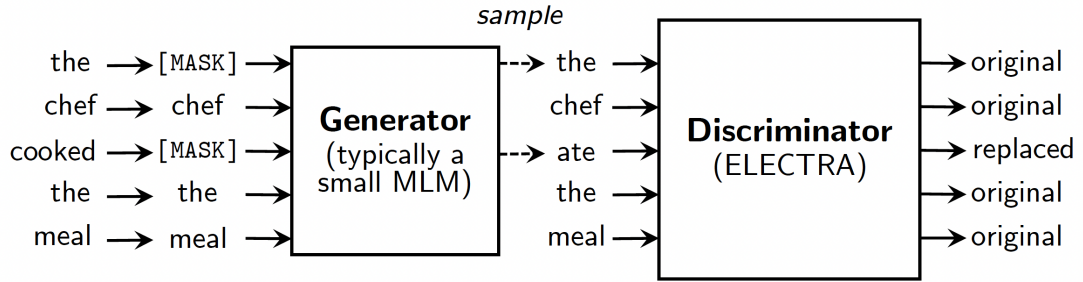


Figure 7: Electra - Architecture Overview (Clark et al., 2020)

function has been applied to it. The Softmax function applied to a model's score vector v of vocabulary size n is calculated row wise, for example for the i th entry

$$\text{softmax}(v_i) = \frac{e^{v_i}}{\sum_{k=1}^n e^{v_k}}. \quad (14)$$

Since this function squashes all the last layer's activations to the interval between 0 and 1 and because naturally all of these squashed activations add up to one, the output can then be interpreted as a probability distribution over the tokens because every entry represents one of the tokens contained in the vocabulary (Braga-Neto, 2020 p. 131). Therefore, the output of a masked language model is a discrete probability distribution over the masked token. The most interesting part of this token distribution is of course the probability assigned to correct token of the object label a model was supposed to retrieve, which the all models are trained to optimize.

2.2.2 Accuracy

In order to calculate the accuracy of a masked PLM, one usually reduces the probability distribution the model outputs to the single index that has the highest assigned probability. But, especially when dealing with natural language processing and PLMs, answers are not always unambiguous and more than one token can be interpreted as a correct result for the mask. The accuracy is therefore given according to a parameter, called k in this work, which describes how many of the most probable token indices were considered. Finally, the accuracy is calculated in this rank-based approach meaning if the correct token is among the top k tokens the answer is correct otherwise it is considered wrong, irrespective of the probability actually assigned to the correct token by the model. In order to denote how many tokens were considered when calculating the accuracy of a model on a data source, we will always speak of the $k@1$ or the $k@10$ accuracy.

2.3 PLMs as Knowledge Bases

The construction and maintenance of large knowledge bases has been the focus of many computer scientists in the last decades. They allow computers to access and utilize

knowledge automatically in programs. But there are two major drawbacks to the current approaches. First in order to use them, specifically structured queries must be used which are as rigid as most programming languages and especially the conversion from human inquiries made to programs into structured queries posed to knowledge bases is a separate problem. Secondly because so much knowledge is added in every conceivable domain in every second there must be automated ways to update and append knowledge bases regularly. If the unsupervised training of neural networks could result in language models that can understand most of human language, like BERT, and be used as knowledge bases with comparative performance, this would obviously represent a huge leap forward. The ease of maintaining their knowledge through training as well their capability of answering very flexible queries could mean the end of traditional knowledge bases, knowledge graphs and their contemporaries. This leads to asking what do language models know.

2.3.1 Knowledge of PLMs

According to Petroni et al. (2019) PLMs such as BERT work surprisingly well as knowledge bases out of the box when compared to supervised baselines. They can also act as question answering tools without further training, but the performance strongly varies regarding the kind of knowledge that is asked for. Compared to traditional knowledge bases which require large annotated data sets with fixed schemata and complex modular pipelines applying many techniques like entity extraction, co-reference resolution, entity linking and relation extraction, PLMs can be trained in a simple and unsupervised manner simply by training them on a given corpus with the right properties. In order to assess and compare language models as knowledge bases, we must first define the term knowledge and describe how one could measure the knowledge inherent in a PLM. There are two clusters of approaches to these questions. We can define knowledge as knowing the answer to a question and assess this knowledge by simply asking and checking the answer. The other definition is based on facts that are represented as relational triplets centered with predefined relations. The knowledge can be assessed converting such triples into so called cloze-statements where the object is masked and needs to be predicted. Petroni et al. (2019) give the following example (Dante, born-in, Florence) which can be checked by posing the following cloze-statement to the language model and checking the most probable predictions for the masked word: "Dante was born in [MASK]". The difference lies in the fact that a question is always constructed with an answer in mind and a cloze-statement can be any statement containing the subject, object and relation of the triplet.

2.3.2 LAMA Probe

The LAMA, Language Model Understanding, probe is a framework to test the factual and commonsense knowledge in language models devised by Petroni et al. (2019) which takes both approaches described above into account, meaning that a PLM is tested with subject-relation-object triples and question-answer pairs. In order to assess the knowledge of the PLM under investigation, it is confronted with triplets from the following data sets

- Google-RE (Google, 2013),
- T-REx (Petroni et al., 2019),
- ConceptNet (Speer, Havasi, et al., 2012).

and question-answer-pairs from SQuAD (Rajpurkar et al., 2016) as a data source. Google-RE is a corpus that is composed of about 60,000 facts categorized in five relations which were manually extracted from Wikipedia. The T-REx knowledge source is based on the T-REx data set (Elsahar et al., 2018) and it is also a subset of facts from Wikipedia organized in 41 Wikidata relations that contain up to 1,000 facts per relation. ConceptNet is a multilingual knowledge base, that was initially built on top of Open Mind Common Sense (OMCS) sentences and is organized into 16 different relations of triplets that have actual sentences attached. For the LAMA probe, only the English sentences of ConceptNet were used. SQuAD is the only question-answer-pair data source used in the LAMA probe, and 305 context-insensitive questions with single token answers were chosen from the source and cloze-statements were constructed for these question answer pairs. Jiang et al. (2020) argue that querying language models with manually created prompts just provides a lower bound on its actual knowledge because a language model's answer to a prompt is highly dependent on how the question is posed. In order to provide an answer that is more than just a lower bound to the question of what a PLM knows, they propose learning a new and entirely separate prompt model. In this endeavor, Jiang et al. (2020) take the fundamental position that if PLMs can correctly complete a fact for most prompts based on it, then one can state that the language models knows this fact. We will refer to the task of completing such a fact as slot filling.

2.3.3 Engineering Prompts

In order to reach a more precise answer to the question of the amount of knowledge contained within a PLM, it is important to pose the right questions. After all, if a language model can not elicit the correct word from one prompt but can when using another, does the language model not know the fact?

2.3.3.1 Manual Prompts

Petroni et al. (2019) do not only develop the LAMA probe as benchmark for testing the factual knowledge of PLMs, they also perform tests with several language models, such as ELMO (Peters et al., 2018) and Bert (Devlin et al., 2018) among others. In order to query these models on the factual knowledge represented by the triplets contained in the data sources, they manually constructed templates for each relation that would generate a cloze-statement for each triplet. Manually constructed means the authors handcraft sentence templates based on the relation where a triplet's subject and object label can be deployed. An example for the place-of-birth template is the following.

[subject label] was born in [object label]

In order to query the model, the subject label is actually deployed, and the object label is masked. Table 1 shows an overview of the results Petroni et al. (2019) achieve on the LAMA probe with manual prompts.

2.3.3.2 LPAQA Prompts

Jiang et al. (2020) build upon the work of Petroni et al. (2019) by working under the assumption that the fact that a language model cannot produce the correct answer for a given cloze-statement does not necessarily mean that the language model does not know the fact. In fact, Jiang et al. (2020) imply that one can state a language model knows a fact related to a triple if any prompt can be found that elicits a triples correct object label for the masked statement. They therefore propose to optimize prompts to tighten the lower bound of a PLMs knowledge found by Petroni et al. (2019). Jiang et al. (2020) use two different approaches in order to optimize prompts, mining-based generation and paraphrase-based generation. Mining-based generation means extracting prompts from a relevant text corpus, in the case of the LAMA probe for example they use Wikipedia. Following the observation that a chain of words found within the text corpus containing the subject and object label of a triple often describes the respective relation. These middle words can be extracted as prompts directly from the text corpus. The second extraction approach is based on dependency parsing. Using this method, a template is identified as the shortest dependency path between the subject label and object label and then using the phrase spanning from the leftmost word to the rightmost word in the dependency path as a prompt. Paraphrase-based generation can automatically be achieved by using back-translation. Back-translation describes the following process, a sentence that is supposed to be paraphrased is first translated into another language and then translated back into the original language. The later translation generates a set of translation candidates that can be used as paraphrased prompts. In order to rank these translation candidates, Jiang et al. (2020) use the round trip probability of the candidates, meaning the probability that they would be translated back into the exact original sentence. Additionally, Jiang et al. (2020) propose using ensemble methods to actually perform the prompt posing. Thus, they develop a prompt model where each relational prompt posed to a language model is fanned out into multiple prompts using either prompts that were mined from the text for the specific relation or paraphrased automatically. In order to construct this prompt model, one needs to employ a training set similar to the actual data source. From the set of answers the language model returns for these prompts, a combined answer is generated. To optimize the combination, the prompt model is trained to learn weights for each of the fanned prompts in order to maximize the influence of useful variants of each prompt. Table 1 shows their results, and we can see that their approach was able to improve the accuracy of the fact retrieval by three percentage points in total. However, because of their use of training data it is somewhat doubtful if their results can truly be interpreted as a lower bound on a PLMs inherent knowledge, as will be discussed in later sections.

2.3.3.3 Autoprompt

Shin et al. (2020) build on the ideas of their predecessors in order to further improve accuracy of fact retrieval from PLMs using prompts that are automatically generated. Their main idea is to learn a number of trigger tokens that elicit an optimal response from the language model under test. More concretely, an AutoPrompt prompt is constructed from three parts. The original subject label, a fixed number of trigger tokens and the mask. The trigger tokens are shared for all prompts within a relation, and thus effectively serve as a prompt template. The trigger tokens are learned from a training data set by first being initialized as masks and subsequently iteratively updated with a gradient based approach to maximize the label likelihood over batches of training examples. It is important to note that the training examples may not overlap with the test set, and the trigger tokens must also not overlap with the token of the object label that is supposed to be elicited from them model. The amount of trigger tokens is a hyperparameter of their statistical model. Table 1 shows that Shin et al. (2020) are able to significantly outperform the original manually crafted LAMA prompts as well as the ensemble-based and mined LPAQA prompts for the LAMA probe benchmark. Shin et al. (2020) do not only apply this method of automatic prompt generation to fact retrieval, but also employ it for fine-tuning tasks with noteworthy results.

2.3.3.4 Optiprompt

Again building upon the ideas of AutoPrompt Zhong et al. (2021) introduce OptiPrompt which continues the idea of using trigger tokens, for which there is only a discrete search space, to trigger embedding vectors in the embedding space which is continuous. This main difference between AutoPrompt and OptiPrompt has the regrettable consequence that in comparison to all previous prompts for fact retrieval from PLMs, the OptiPrompt prompts are no longer human-readable. The templates for OptiPrompt prompts are again constructed through optimization of trigger vectors on the same training set that was used by Shin et al. (2020) for AutoPrompt but the trigger embedding vectors, the amount of these vectors and their position is determined by the manual prompts from the LAMA probe. The authors note that they believe good initialization for the trigger vectors to be a necessity for finding a good solution to this challenging non-convex optimization problem. Table 1 shows that the OptiPrompt prompts again outperform AutoPrompt prompts and their predecessors and delivers the best performance on the LAMA probe.

2.3.3.5 Optimizing Prompts

Furthermore, Zhong et al. (2021) discuss the impact that prompt models have on eliciting facts from PLMs and question the accuracy of the results obtained from optimizing prompts by using a training set. They ask if further knowledge can be learned from the training set and thus be encoded into the prompt model, and thus leading the combination of language and prompt model to overestimate the actual knowledge of the dataset. Zhong et al. (2021) that prompt models such as LPAQA, AutoPrompt and OptiPrompt in fact exploit regularities and correlations in the underlying training set by first showing that these regularities exist and second by revealing that these prompt models actually exploit them. Regularities in the training set that can be exploited, for example,

are dominant object labels. The authors trained two probabilistic models, a class prior and a Naive Bayes model, on the training set and reached accuracy levels of over 20% on the LAMA Probe, thus demonstrating that the training sets themselves contain useful information for fact retrieval. This is not surprising, considering the fact that some of the LAMA Probe’s relations have one or two object labels for half of the relation triplets. Zhong et al. (2021) demonstrate that the prompt models indeed exploit these regularities by initializing a BERT with random weights and combining it with the prompt models to again reach accuracy levels up to around 20% percent. This is clear evidence that training prompt models on training data and using it to query language models distorts the accuracy measured on the lama probe.

Method	1:1	N:1	N:M	Total
LAMA	68.0	32.4	24.7	31.1
LPAQA	65.0	35.9	27.9	34.1
AutoPrompt	58.0	46.5	34.0	42.2
OptiPrompt	59.6	54.1	40.1	48.6

Table 1: Micro-averaged k@1 Accuracy in Percent of Bert-Base-Cased averaged on the LAMA Probe

2.3.4 Related Work

In related work, Cao et al. (2021) and Zhong et al. (2021) demonstrate notable limitations of the LAMA probe and its successors. Cao et al. (2021) strongly question the viability of PLMs to serve as reliable factual knowledge bases. They show that prompt-based retrieval is severely prompt-biased, which means the prompt has an outsized impact on the answer distribution. They show this through the construction of an entirely new data source based on Wikidata having the same properties, with highly correlated answer distributions for the same prompts. This suggests that the superiority of prompts is caused by a better fit for the answer distribution, rather than some inherent retrieval ability. Furthermore, they find that illustrative cases improve slot filling performance mainly because they offer entity type guidance and golden answer leakage. These results can be summarized as stating the process that leads to model an answer is closer to guessing, although educated guessing, than providing inherent knowledge saved deep within the model. Zhong et al. (2021) show that much of the improvement of manual prompts gained by the successors of Lama stems not from the LMMs originally training phase, in other words from the original fact learning phase, but from the subsequent prompt training. In order to address these shortcomings Kalo and Fichtel (2022) introduce KAMEL, Knowledge Analysis with Multitoken Entities. Conceived as a superior PLM knowledge base performance benchmark, it is a larger data source that covers a broader range of knowledge, probes for single- and multi-token entities as well as contains facts with literal values. They find that the performances of LLMs drastically decrease when applied to KAMEL instead of LAMA. They conclude that PLMs are as of now not even close to being able to memorize all varieties of relational knowledge that is usually stored in knowledge graphs. These results are contrasted by the work of Heinzerling and Inui (2020) who offer a proof of concept and conclude that language models can indeed serve

as knowledge bases. They propose and compare three methods to extend LMs to KBs containing millions of entities.

- Symbolic representation by extending the LM vocabulary with entries for all entities
- Surface form representation, each entity is represented by its subword-encoded canonical name, which is stored and queried by extending the LM with a sequence decoder
- Continuous representation, where each entity is represented as an embedding

Each of the three proposed extension methods comes with its own set of tradeoffs between computational cost and entity-linked data requirements.

3 Grammar and Syntax Structure

In this chapter we will define the basic vocabulary and introduce the important concepts of linguistics and grammar that are needed to answer the guiding questions of this work, the sentence typology and the transformations passivization and nominalization.

3.1 Fundamentals of Linguistics

In this section, we will introduce the basic linguistic concepts and introduce the distinctions whose effects on the answers of language models we will study in this work.

3.1.1 Sentence and Clause

According to Huddleston (1984) syntax is concerned with the way words combine to form sentences and therefore the sentence is the largest piece of language forming a syntactic construction. Everything larger than a sentence would thus be classified as a text. Those two categories themselves differ from one another by the property that a sentence is governed by rules, which dictate how words may be combined to construct a sentence, but there are no such rules governing how sentences may be combined to form a text, apart from a disputable distinction between good and bad writing.

3.1.2 Verbs and Inflection

English is classified as a slightly inflected language because verb forms vary marginally with conjugation. Nonetheless, verbs may inflect for grammatical categories such as person, number, gender, case, tense, active versus passive and many more aspects (Passer, n.d.).

3.2 Sentence Typology

Huddleston (1984) in his classic textbook on English grammar describes a typology of sentences that is composed of three categories. According to this categorization, sentences are either

- simple
- compound or
- complex.

Huddleston (1984) differentiates these sentence types by the amount of main and dependent clauses it is composed of and gives the following definitions.

Definition 1 *A simple sentence is defined as a sentence that contains only one main clause.*

Definition 2 *A compound sentence is a sentence that contains two or more main clauses.*

Definition 3 *A complex sentence contains two or more clauses, but must contain at least one subordinate clause.*

Huddleston (1984) does not make a distinction between a sentence that has but one main but one subordinate clause and a sentence that has more than one main clause. Following his typology, both thus constructed sentences would be classified as complex. Since this work is ultimately not just interested in Linguistics but in the interplay between sentence structures and PLMs, we regard such sentences as sufficiently different to justify a further distinction between merely complex sentences and sentences that are compound and complex. Therefore, in this work, we will work with the following sentence typology based on the following four classes of sentences.

1. simple
2. compound
3. complex
4. compound-complex

Thus, we need to change the definition of the complex sentence and add a new definition for the compound-complex sentence.

Definition 4 *A complex sentence contains one main clause and one subordinate clause.*

Definition 5 *A compound-complex sentence contains two main clauses and one subordinate clause.*

An example of a simple sentence would be "The native language of Johann Wolfgang von Goethe is German". This information can also be expressed in a compound sentence the following way, "Johann Wolfgang von Goethe speaks German, and it is also his native language". Transformed into a complex sentence, it would be "Johann Wolfgang von Goethe has a native language, which is German" and finally expressed as a compound-complex sentence "Johann Wolfgang von Goethe speaks German, and he speaks it well because it is his native language".

3.3 Passivization

Passivization is another linguistic transformation that will be investigated in this work. Passivization as well as its counterpart, the activation is the process of switching the subject and the object of a sentence, changing the verb's inflection and if necessary adding or subtraction a by clause according to Huddleston (1984). For example, the sentence "Cristiano Ronaldo plays the striker position" can be transformed through passivization into the sentence "The striker Position was played by Cristiano Ronaldo".

3.4 Nominalization

Nominalization is, as the name suggests, the transformation of a phrase or sentence, which turns a verb or a verb-based part into a noun or a noun-based part of a sentence. Huddleston (1984) defines nominalization as the grammatical process resulting in forms bearing a resemblance to nouns or noun phrases - a greater resemblance than the form from which they are derived. This definition also encompasses sentences that do not actually contain nouns. Therefore, Huddleston (1984) differentiates between lesser and greater degrees of nominalization. A nominalization of greater degree always actually results in a transformed phrase or sentence that contains a noun that was not present in the original phrase or sentence before the transformation. A nominalization of lesser degree is a transformation into a more noun like phrase or sentence that does not actually contain any additional nouns. In this work, the use of the term nominalization will always be referring to a nominalization of greater degree. Colloquially speaking, nominalization turns a verb into a noun. An example of nominalization included in later experiments is the original sentence "Apple develops the operating system iOS" nominalized into "The operating system iOS is a development of Apple". The nominalized verb in this example is "develop" who's nominalized form is "development".

4 Experiments

In this chapter we will describe, analyze and interpret the experiments designed to investigate the influence of syntactical structure on fact retrieval with language models. First we will compare the accuracy of and token distributions of the different sentence types from the sentence typology followed by the comparison of active, passive and nominalized sentences. Afterwards we continue with experiments designed to test hypothesis formed on the basis of the first experiments.

4.1 T-REx Data Source

The T-REx data source is a knowledge base derived from Wikidata triples, which was introduced by Elsahar et al. (2018). The data is organized in 41 different Wikidata relations, with an average of 830 triples sub-sampled per relation. The 41 relations cover all possible cardinality types $1 : 1$, $N : 1$ and $N : M$. Table 2 provides a general overview over the data set, Table 3 describes all relations in detail, and Table 4 shows examples for the objects and subject for each relation.

Cardinality	No. of triples	No. of relations	average No. of triples per relation
Total	34,039	41	830.2
1:1	937	2	468.5
N:1	20,928	24	872.0
N:M	12,174	15	811.6

Table 2: T-REx Meta Information

4.2 Sentence Typology

In this section, we explore fact retrieval performance differences of several PLMs caused by different syntactical structures. The exploration procedure is based on comparing the overall accuracy for the four different sentence types of the sentence typology introduced in the previous chapter. Therefore, we first must create prompt templates for each of the four sentence types and each of the 41 relations included in the T-REx data source.

4.2.1 Templates

For each of the 41 relations of the T-REx data source, we create a set of four templates, one sentence template for each of the sentence types of the sentence typology. The sentence templates were manually created with the intent of carrying roughly the same semantic content or information as far as that was possible. It would be much easier to construct sentence type templates without this restriction and the sentences would often sound more natural, but it was important to keep the semantic content equivalent as much as possible in order to really observe the effect of the changing syntactical structure. The following three examples shown in Table 51 give an impression of these templates. The full set of templates is available in the Appendix and online.

4.2.2 Experimental Setup

The experiment is conducted by querying the PLMs with four different sentences, each based on each of the types described in the sentence typology for all triplets of all 41 relations for the accuracy levels $k@1$ and $k@10$. The subject labels are injected into the templates the object labels are masked, and the result is recorded as 1 if the token for the object label is among the top k predicted tokens, otherwise it is labeled as 0. The accuracy is calculated in three ways, first we calculate the total average accuracy over all

Relation	Name	Cardinality	size
P17	country	N:1	930
P19	place of birth	N:1	944
P20	place of death	N:1	953
P27	country of citizenship	N:M	966
P30	continent	N:1	975
P31	instance of	N:1	922
P36	capital	1:1	703
P37	official language	N:1	966
P39	position held	N:M	892
P47	shares border with	N:M	922
P101	field of work	N:M	696
P103	native language	N:1	977
P106	occupation	N:M	958
P108	employer	N:M	383
P127	owned by	N:1	687
P131	located in the administrative territorial entity	N:1	881
P136	genre	N:1	931
P138	named after	N:1	645
P140	religion or worldview	N:1	473
P159	headquarters location	N:1	967
P176	manufacturer	N:1	982
P178	developer	N:M	592
P190	twinned administrative body	N:M	995
P264	record label	N:1	429
P276	location	N:1	959
P279	subclass of	N:1	964
P361	part of	N:1	932
P364	original language of film or TV show	N:1	856
P407	language of work or name	N:1	877
P413	position played on team / specialty	N:1	952
P449	original broadcaster	N:1	881
P463	member of	N:M	225
P495	country of origin	N:1	909
P527	has part or parts	N:M	976
P530	diplomatic relation	N:M	996
P740	location of formation	N:1	936
P937	work location	N:M	954
P1001	applies to jurisdiction	N:M	701
P1303	instrument	N:M	949
P1376	capital of	1:1	234
P1412	languages spoken, written or signed	N:M	969

Table 3: T-REx Data Set relations with names and cardinalities (Elsahar et al., 2018)

Relation	Name	Subject Example	Object Example
P17	country	Germany	Eibenstock
P19	place of birth	Alexandra	Allan Peiper
P20	place of death	Iraq	Akihiko Saito
P27	country of citizenship	Brazil	Rubens Barrichello
P30	continent	Antarctica	Lavoisier Island
P31	instance of	castle	Tantallon Castle
P36	capital	Baku	Azerbaijan
P37	official language	Urdu	Azad Kashmir
P39	position held	archbishop	Bernard Francis Law
P47	shares border with	Palermo	Monreale
P101	field of work	logic	Alan Turing
P103	native language	French	Louis Jules Trochu
P106	occupation	actor	Shreela Ghosh
P108	employer	Apple	Steve Jobs
P127	owned by	Apple	iWork
P131	located in the administrative territory	Northamptonshire	A605 road
P136	genre	opera	John Cage
P138	named after	Aristotle	aristotelianism
P140	religion or worldview	Buddhism	Gautama Buddha
P159	headquarters location	Colchester	Chandos Records
P176	manufacturer	Fiat	Fiat Multipla
P178	developer	Apple	MessagePad
P190	twinned administrative body	Mainz	Valencia
P264	record label	Decca	Buddy Holly
P276	location	Paris	Eiffel Tower
P279	subclass of	galaxy	lenticular galaxy
P361	part of	Asia	South Asia
P364	original language of film or TV show	Russian	The Scarlet Flower
P407	language of work or name	German	Die Zeit
P413	position played on team / speciality	midfielder	Koke
P449	original broadcaster	NBC	The Rousters
P463	member of	NATO	Albania
P495	country of origin	Italy	Soppressata
P527	has part or parts	carbon	pectin
P530	diplomatic relation	Germany	Paraguay
P740	location of formation	Baltimore	Pianos Become the Teeth
P937	work location	London	Howard Florey
P1001	applies to jurisdiction	France	president of the Republic
P1303	instrument	saxophone	Kenny G
P1376	capital of	Alberta	Edmonton
P1412	languages spoken, written or signed	Italian	Iginio Ugo Tarchetti

Table 4: T-REx Data Set relations with examples (Elsahar et al., 2018)

typological-transformation	Template
<i>simple</i>	The capital of [S] is [O]. [S] is affiliated with the religion called [O]. [S] maintains diplomatic relations with [O].
<i>compound</i>	[S] is a country and its capital is [O]. [O] is a religion or a worldview and [S] is affiliated with it. [S] maintains diplomatic relations with countries and [O] is one of them.
<i>complex</i>	[S] is the country, who's capital is [O]. [S] is affiliated with [O] which is a religion or a worldview. [S] is a country that maintains diplomatic relations with [O].
<i>compound-complex</i>	[O] is a city, and it is the city that is the capital of [S]. Many people are associated with a religion or a worldview and [S] is affiliated with one of these, which is [O] specifically. [S] is a country that maintains diplomatic relations with many others, and [O] is one of them.

Table 5: Templates for P36, 'capital of' (1:1), P140, 'religion or worldview' (N:1) and P530, 'diplomatic relation' (M:N)

triplets for all relations, second we calculate the average accuracy of all triplets belonging to the same relation cardinality, and finally we calculate the average performance for each relation individually. We calculate all average separately.

4.2.3 Results

This section describes the results of the experiment for the sentence typology for all PLMs. These results will clearly show that the models under investigation, at least those yielding acceptable results, clearly favor one syntactic structure over the others. For the analysis of the sentence typology, the favored syntactical structure will be the simple sentence type. In this first level analysis of the results, we first of all show the results and scrutinize them from different perspectives in order to explain the dominance of the simple sentence for the fact retrieval task for these models. First, we undertake a model comparison in which all models will be compared to one another regarding their accuracy performance on the T-REx data source for all four different types of the sentence typology. We will analyze the structure of these results, find explanations for them, and attempt to identify other interesting patterns. After this global model comparison, we will choose the models yielding the most interesting results and analyze the results for different metrics before comparing the results for the individual relations. Finally, we will analyze the score values of these models.

4.2.3.1 Model Comparison

We will begin this model comparison by showing different accuracy metric results and analyzing these for structures and interesting patterns; first in a global comparison before

categorizing the models into groups and looking at them separately. We will establish that all models strongly prefer the single sentence type, meaning that the prompts generated by using this syntactical structure generate the highest accuracy results across the board for all models under investigation, even those with poor general results. Additionally, we will discover that there is a clear disadvantage for uncased versions of the models studied inherent to the TRE-x data source since many of the facts to retrieve are names of people, countries, companies or other comparable entities. Uncased versions have clear disadvantages, differentiating the company Apple from the fruit apple, for example. We will take a separate look at this issue later in this chapter.

4.2.3.1.1 Global Model Comparison

Table 6 shows the k@1 accuracy for each model in percent for all four sentence types. Generally the values of the accuracy results appear quite low, the best performing model could not even predict a third of the masked tokens correctly. Taking into account however that more than a third of the relations are of cardinality type $N : M$, this result seems surprisingly high. Generally, most of the better results hover around 15 to 20 percent. A central conclusion can be drawn forthwith; for all models under investigation, the average k@1 accuracy varies significantly between the different sentence types. This indicates that grammatical structure has a notable effect on a model’s ability to retrieve a fact in all models under investigation.

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	30.221	16.275	16.989	17.994
BERT-base-cased	28.185	12.400	12.953	15.767
BERT-base-multilingual-cased	19.989	13.399	13.388	10.967
BERT-large-uncased	3.381	1.102	1.199	0.467
BERT-base-uncased	3.070	0.752	1.542	0.767
BERT-base-multilingual-uncased	3.499	0.602	1.830	0.388
electra-base-uncased	1.790	0.750	0.340	0.170
SenseBERT-base-uncased	0.43	0.14	0.01	0.00

Table 6: Sentence Typology Model Comparison for k@1 accuracy in percent

Considering emerging patterns in the results shown in this table, Table 6, three conclusions are immediately obvious. First, the model BERT-large-cased outperforms all other models on all four sentence types within the sentence typology by at least two and at most four percentage points. The second apparent observation is the fact that there is a large chasm in accuracy results between cased and uncased models, and the accuracy of uncased models is comparatively and exceedingly low. The third observation shows that every model has a higher prediction accuracy when confronted with the simple sentence type compared to the other three types. Furthermore, the differences between the k@1 accuracy for compound and complex models are close-range and the k@1 accuracy for the compound-complex type seems to be the outlier. Lowering the difficulty threshold for the models a little bit, we move on to consider the top ten among the predicted tokens

to calculate the accuracy of the model for this metric. Table 7 shows the k@10 accuracy for each model in percent for all four sentence types. The first thing to note is that the resulting accuracy values for k@10 are significantly higher than for the k@1 accuracy. The best model, again BERT-large-cased, is now able to predict almost 60% of tokens correctly, meaning is able to retrieve the correct fact a little less than two thirds of the time. Even the uncased models, whose k@1 accuracy performance is abysmal, improves dramatically while still remaining significantly lower than even the worst k@1 accuracy values reached by the cased version for k@1.

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	58.483	41.755	43.738	42.816
BERT-base-cased	57.737	37.319	39.816	38.961
BERT-base-multilingual-cased	39.293	33.359	30.677	32.169
BERT-large-uncased	11.337	6.513	6.384	5.135
BERT-base-uncased	9.33	6.778	6.067	5.361
BERT-base-multilingual-uncased	9.404	4.307	5.285	3.543
electra-base-uncased	5.452	3.896	3.126	3.960
SenseBERT-base-uncased	2.070	0.740	0.460	0.210

Table 7: Model Comparison for k@10 accuracy in percent

Moreover, we can directly see that all the three observations regarding the k@1 accuracy also apply to the k@10 accuracy results. The first observation from the k@1 accuracy shown in Table 6 holds for the k@10 accuracy because the cased version of the large BERT-model outperforms all other models. The second one also holds, since there is again a big difference between the cased and the uncased BERT-based models, in this case an even larger discrepancy in every regard. Also, the final observation applies again the k@10 accuracy is higher for the simple sentences for every single model. As we have discovered so far from the results for the k@1 and k@10 average accuracy results, the accuracy values dramatically improve when considering the top ten predicted tokens instead of just the token with the absolute highest score, but the structure and the patterns of the results do not change. These patterns do not exactly hold when we move from k@10 to k@100 as Table 8 shows. The result accuracy values have again improved, reaching up to more than 85% but BERT-large-cased is not as dominant as in the previous results because for simple sentences the large BERT-based model is outperformed by the cased version of BERT-base. Nevertheless, there is still a large difference between cased and uncased versions of the BERT-based models.

In the next paragraph we will ask what information, apart from the rising performance, be gained by analyzing the different accuracy results for k@1, k@10 and k@100 average accuracy results for any single model. But for comparing the models to each other, it seems sufficient to focus on a single one of these metrics. Therefore, we will focus on just one of the three choices in accuracy metric results in the model comparison going forward. Questioning the impact of the grammatical structure on fact retrieval accuracy as measured in these experiments, we observe that especially the relative pairwise discrepancy between the accuracy values decreases as the number of tokens that are considered, specified by the k-value, increases. But the differences between the simple sentence type

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	83.487	67.023	69.124	65.607
BERT-base-cased	85.279	61.344	65.877	61.65
BERT-base-multilingual-cased	62.684	52.131	57.226	53.774
BERT-large-uncased	19.501	14.965	14.381	12.571
BERT-base-uncased	19.281	14.057	13.532	11.243
BERT-base-multilingual-uncased	18.385	11.325	11.152	9.789
electra-base-uncased	12.495	9.257	8.047	9.319

Table 8: Model Comparison for k@100 accuracy in percent

and the other sentence types remains most pronounced. This indicates that most models under investigation are able to retrieve facts better for simple sentences than for more complicated grammatical structures, similar to most humans. An additional finding of interest is the fact that the smallest accuracy discrepancy for most models can be found comparing the compound to the complex sentence type, especially for the cased BERT-base models, which are the best performers in this experiment, the accuracy differences are minuscule for all three k values. From the global model comparison, we were able to identify two distinct groups based on different accuracy result levels for all three k values. There is a group of top performers and a group of low performers. The grouping of models in these categories is shown in Table 9.

Category	Models
Top Performers	BERT-large-cased BERT-base-cased BERT-base-multilingual-cased
Low Performers	BERT-large-uncased BERT-base-uncased BERT-base-multilingual-uncased electra-base-uncased

Table 9: Model Comparison Categorization

4.2.3.1.2 Top Performers

Let us take a look at the category of Top Performers first. This group is composed of the cased versions of the three BERT-based models. This is and will remain their biggest similarity, they are models trained on a cased vocabulary. Since the T-REx data source distinguishes between three different kinds of relations based on the cardinality of these relations, it is of interest to know how these models perform for each of these relations separately. For relations with cardinality 1:1 and 1:N there is only a single correct answer, but for the N:M relations there are numerous correct answers. Therefore, the low k@1 accuracy values for all models are somewhat misleading because a large portion of the relations in the T-REx data source are of cardinality M:N. Table 10 shows the k@1 accuracy for the models within the Top Performers group for the 1:1 cardinality relations. Since

there is only a single correct answer in the 1:1 relations, comparing them using the k@1 accuracy metric seems most appropriate.

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	70.651	69.477	67.556	58.164
BERT-base-cased	65.101	64.461	63.501	48.132
BERT-base-multilingual-cased	47.385	39.701	46.745	19.851

Table 10: Sentence Typology Model Comparison for Top Performers with 1:1 relations and k@1 accuracy

We can clearly see that this table does not show many additional patterns that are not visible in the global comparison for all relations because BERT-base-large is still the best performing model among the models and the models achieved the best accuracy values when prompted with the simple sentence type sentences. But the accuracy levels are much higher for these relations than for the global comparison, and the differences in accuracy levels between the compound and the complex sentence type are a little bit starker. This could be caused by the low overall measurements because with only two relations the 1:1 cardinality relations amount to less than 1,000 prompts over all. The most striking difference is that the compound-complex sentence type is outperformed by all other types for the 1:1 cardinality relations in all three models by a significant margin. This is in contrast to the global results, where the compound-complex sentence type registers higher accuracy values for BERT-large-cased and BERT-base-cased. Moving on to the 1:N cardinality relations for the Top Performer group, using again the k@1 accuracy metric because there is only a single correct token, we get the following results shown in Table 11. The k@1 accuracy levels are in congruence with their global counterparts, but a little higher.

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	35.073	18.353	18.411	20.126
BERT-base-cased	31.962	14.512	16.523	19.481
BERT-base-multilingual-cased	23.347	15.916	16.418	14.187

Table 11: Sentence Typology Model Comparison for Top Performers N:1 k@1 accuracy

The largest model, BERT-large-cased, performs better than the others again and the best accuracy values are measured for the simple sentence type. Compound and complex sentences are outperformed by their combination, again as in the global analysis. Table 12 shows the k@10 accuracy for Top Performers just for the N:M cardinality relations. We choose the k@10 accuracy metric because in such a relation, there are many different correct answers.

Applying these filters, the general structure of the results does not change by much, except in two ways, that are interesting to note. On one hand, the k@10 accuracy results for just the N:M cardinality relations are more than ten percentage points lower than the global k@10 results for every model in the Top Performers group. Secondly, the pair wise accuracy differences between the three non-simple sentence types is much bigger than in

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	43.568	35.781	32.224	35.412
BERT-base-cased	44.726	32.249	29.046	32.372
BERT-base-multilingual-cased	30.853	27.559	20.289	22.31

Table 12: Sentence Typology Model Comparison for Top Performers N:M k@10 accuracy

the other results. Since it is not clear what the maximum number of correct answers in all the M:N cardinality relations is, it is prudent to take a further look at the k@100 results for these relations, which can be seen in Table 13. But as we see, this perspective does not yield any further insight than the global k@100 accuracy results did.

Model	Simple	Compound	Complex	Compound-complex
BERT-large-cased	75.456	63.496	56.547	58.732
BERT-base-cased	83.473	58.666	57.467	54.386
BERT-base-multilingual-cased	61.138	49.384	53.959	48.283

Table 13: Sentence Typology Model Comparison N:M for Top Performers k@100 accuracy

In conclusion, the models belonging to the Top Performers group outperform the global accuracy values for 1:1 and 1:N cardinality relations and under-perform for the M:N cardinality relations. There is always a clear hierarchy of accuracy performance with the largest model BERT-large at the top followed by its smaller version BERT-base and then followed by the multilingual model. Which is completely in line with what one would predict if one would just try to predict these performances, knowing nothing but the size of the networks and vocabularies involved.

4.2.3.1.3 Low Performers

The Low Performer group consists of the models of the Top Performer group in the uncased version plus the models where no cased version is publicly available as of the time of this writing, namely SenseBERT-base-uncased and electra-base-uncased. The low performers are of limited value to this investigation because we suspect and show later in this work that the low accuracy levels are a phenomenon best explained by the fact the T-REx data set contains too many named entities and set the uncased models at a disadvantage because they cannot differentiate them out of the box from other entities with the same name. In later sections of this chapter, we filter out named entities and demonstrate that this causes the discrepancies between cased and uncased models to shrink significantly.

4.2.3.2 Metric Comparison

In this metric comparison we will exclude all the models that are categorized as low performers and focus on the top performing models, namely BERT-large-cased, BERT-

base-cased, BERT-base-multilingual-cased. We will look at the accuracy results for every one of the three chosen models and analyze the results. These results will again clearly demonstrate that the syntactical structure of a sentence matters greatly to the results because all models clearly favor the simple sentence type over the other types.

4.2.3.2.1 BERT-Large-Cased

Table 14 shows the results for the different metrics for BERT-large-cased. We recognize the familiar structure of the results, the simple sentence type outperforms all other sentence types by a significant margin. We analyze the differences between the sentence types for this model in this paragraph.

Metric	Simple	Compound	Complex	Compound-complex
k@1 accuracy	30.221	16.275	16.989	17.994
k@10 accuracy	58.483	41.755	43.738	42.816
k@100 accuracy	83.487	67.023	69.124	65.607

Table 14: Sentence Typology Metric Comparison BERT-large-cased

Table 15 shows the differences between the average accuracy values for the simple sentence and the other sentence type sentences. As we can see, the differences are surprisingly constant for all accuracy metrics. All values lie within 13 and 18 percentage points. Which means the difference does not increase as more tokens are considered as valid answers. This could indicate that there is a constant share of T-REx triples, roughly the size of 13 percent, for which the models are only able to retrieve the correct facts for simple sentence types and not for the other types.

Metric	Compound	Complex	Compound-complex
k@1 accuracy	13.946	13.232	12.227
k@10 accuracy	16.728	14.745	15.667
k@100 accuracy	16.464	14.363	17.880

Table 15: Differences to Simple Results Sentence Typology Metric Comparison BERT-large-cased

Turning away from the differences in average accuracy performance between the simple sentence type and the other sentence types, we now assess the differences among the other sentence types. Table 16 shows the results, excluding the results for the simple sentence type. We can clearly see that they do not establish a strong hierarchy between the different types. The compound-complex class for example has the highest accuracy values for k@1, the second best for k@10 and the worst for k@100. The only discernible new pattern is the fact the complex sentence type always slightly outperforms the compound sentence type.

4.2.3.2.2 BERT-Base-Cased

Metric	Compound	Complex	Compound-complex
k@1 accuracy	16.275	16.989	17.994
k@10 accuracy	41.755	43.738	42.816
k@100 accuracy	67.023	69.124	65.607

Table 16: Sentence Typology Metric Comparison BERT-large-cased without simple sentence type

Table 17 shows the results for the different metrics for BERT-base-cased. Unsurprisingly, we can identify the same pattern as for the larger BERT-based version of the model. The model can predict the correct token or one of the correct tokens with the highest accuracy values for the simple sentence type, again outperforming the other sentences types by a significant margin.

Metric	Simple	Compound	Complex	Compound-complex
k@1 accuracy	28.185	12.400	12.953	15.767
k@10 accuracy	57.737	37.319	39.816	38.961
k@100 accuracy	85.279	61.344	65.877	61.650

Table 17: Sentence Typology Metric Comparison BERT-base-cased complete

Analyzing the differences between the average accuracy values for the simple sentence type and the other three sentence types, see Table 18, we see that the pattern of an almost constant difference does not emerge. Rather, the distance does grow with the number of tokens considered for the evaluation. Although these differences do not rise proportional. The differences for BERT-base are also larger than the difference for BERT-large by up to 8 percentage points.

Metric	Compound	Complex	Compound-complex
k@1 accuracy	15.785	15.232	12.418
k@10 accuracy	20.418	17.921	18.776
k@100 accuracy	23.935	19.402	23.629

Table 18: Differences to Simple Results Sentence Typology Metric Comparison BERT-base-cased

Excluding now, as before, the average accuracy results for the simple sentences we get the results shown in Table 19 we get exactly the same structure as for BERT-base-large because there is again no clear hierarchy between the non-simple sentence types emerging other than the fact that the complex sentence type always outperforms the compound sentence type.

4.2.3.2.3 BERT-Base-Multilingual-Cased

Table 20 shows the results for the different metrics for the multilingual BERT-based model, BERT-base-multilingual-cased. As do the other BERT-based models, so the av-

Metric	Compound	Complex	Compound-complex
k@1 accuracy	12.400	12.953	15.767
k@10 accuracy	37.319	39.816	38.961
k@100 accuracy	61.344	65.877	61.650

Table 19: Sentence Typology Metric Comparison BERT-base-cased complete without simple sentence type

average accuracy results for the multilingual version also demonstrate a significant margin in favor of the simple sentence type.

Metric	Simple	Compound	Complex	Compound-complex
k@1 accuracy	19.989	13.399	13.388	10.967
k@10 accuracy	39.293	33.359	30.677	32.169
k@100 accuracy	62.684	52.131	57.226	53.774

Table 20: Sentence Typology Metric Comparison BERT-base-multilingual-cased

Analyzing again the differences in average accuracy for the simple sentence type to the other sentence types, we get the following table, Table 21. We see the most interesting pattern among the difference tables we have inspected. The differences are closer together in value than for the previous two models, at maximum only separated by less than 6 percentage points. Secondly, they are not strictly rising with the k values as expected. Furthermore, the differences seem to be almost randomly distributed in the table, all the three non-simple sentence types account for a single maximum difference for a single k value.

Metric	Compound	Complex	Compound-complex
k@1 accuracy	6.590	6.601	9.022
k@10 accuracy	5.934	8.616	7.124
k@100 accuracy	10.553	5.458	8.910

Table 21: Differences to Simple Results Sentence Typology Metric Comparison BERT-base-multilingual-cased

The accuracy comparison between the three non-simple sentence types also yields a result that is different from the monolingual counterparts of the BERT-based models. Table 22 shows the results excluding the simple sentence type. We can again not identify a clear hierarchy among the non-simple sentence types based on average accuracy results, but surprisingly the compound sentence type yields better results for the multilingual model than the complex sentence type for the two lower k values.

In conclusion, we can identify a single strong and clear pattern from the metric analysis of the top performing models. The simple sentence types dominate all others in every metric and in every model. Other than that, the compound sentence type mostly produces the worst average accuracy results, certainly worse than the complex for most models for all metrics.

Metric	Compound	Complex	Compound-complex
k@1 accuracy	13.399	13.388	10.967
k@10 accuracy	33.359	30.677	32.169
k@100 accuracy	52.131	57.226	53.774

Table 22: Sentence Typology Metric Comparison BERT-base-multilingual-cased

4.2.3.3 Relation Comparison

We will now take a much closer look at the structure of the average accuracy results by analyzing these results on the level of the three different cardinalities and individual relational levels. Once again, we will focus on the top performing models identified in model comparison and analyzed in the metric comparison. These results will give us a more differentiated picture of the influence of syntactical structure on the average accuracy values by demonstrating that there is a clear difference in accuracy between all four sentence types for some relations, a small difference for other relations and almost everything in between. Thus, these results once again illustrate that these differences between the fact retrieval performances can not easily be explained by superficial factors. We will see that looking at the individual relations or the cardinalities does not deliver a satisfactory explanation for the differences in performance or the reason for the dominance of the simple sentence type.

4.2.3.3.1 BERT-Large-Cased

Additional to the global average accuracy result, Table 23 and Table 24 show the average accuracy results for the four sentence types for each of the cardinality relations for k@1 and k@10 for BERT-large-cased. Both results show that the simple sentence type has the highest accuracy for all three cardinalities. Also in both sets of results the level of accuracy results decreases as the cardinality increases. This phenomenon is to be expected because it is intuitively clear that the difficulty level increases with the amount of possible subjects and especially objects.

Results	Simple	Compound	Complex	Compound-Complex
Total	30.22	16.59	16.08	17.87
1:1	70.65	69.48	67.56	58.16
N:1	35.07	18.36	18.18	20.01
N:M	18.77	8.61	10.65	11.24

Table 23: k@1 Accuracy in Percent of BERT-Large-Cased on the T-REx Data Set

The results are the closest for the cardinality 1:1 and furthest apart for N:M. Furthermore, there is no hierarchy emerging with the exclusion of the simple sentence type because among the 6 rows in both tables every other sentence type reaches the second-highest accuracy result at least once. In general, the greatest drop of in accuracy levels between the simple and the other sentence types occurs for the relations with the N:1 cardinality.

Results	Simple	Compound	Complex	Compound-Complex
Total	58.48	42.18	42.75	43.09
1:1	85.17	84.95	84.31	81.32
N:1	65.96	43.49	48.47	45.40
N:M	43.57	36.62	29.70	36.18

Table 24: k@10 Accuracy in Percent of BERT-Large-Cased on the T-REx Data Set

We will now look at the average k@10 accuracy results for the individual relations. These are demonstrated in Table 25

In order to finally establish a hierarchy based on more than one value, we can count how many times a sentence type generates the top accuracy value for a relation and can come up with the following result shown in Table 26. This is another way of demonstrating the large gap between the performance of the simple prompts versus the others. In this ranking the complex-compound model performs second best, followed by the compound and finally by the complex sentence type.

The relation P413, with label position *played on team or specialty* and cardinality 1:N is an outlier because all three non-simple sentence type are not able to retrieve a single correct token for the prompts and generate a k@10 accuracy of 10%. The same is true for relation P1303, an N:M cardinality relation, labeled *instrument*. Both of these outlier relations have the word *play* in common. One plays a position and one plays an instrument, so the verb play plays an important role in all for all templates of both relations, as the following templates clearly show.

1. simple: <subject label> plays in <object label> position.,
2. compound: <object label> is a position and <subject label> plays in it.,
3. complex: <object label> is the position that <subject label> plays in.,
4. compound-complex: <object label> is one position out of many, and it is the one that <subject label> plays in.

For this relation, the model clearly favors a most simple prompt over other variants. Relation P103 is an interesting relation for many reasons. P103 is the native language relation with cardinality N:1, and it is interesting because firstly the accuracy results for all four sentence types are remarkably high, all reaching more than 96.4% k@10 accuracy and second because all three non-simple sentence types outperform the simple sentence type and lastly because it is the highest accuracy value generated by the compound-complex sentence type considering all relations with 97.5% but still outperformed both by the compound and the complex sentence types on their own. Looking at the templates of this relation,

1. simple: The native language of <subject label> is <object label>.,
2. compound: <subject label> speaks <object label> and it is also his native language.,

Relation	Simple	Compound	Complex	Compound-complex
Total	58.48	42.18	42.75	43.09
P17	67.527	52.688	73.548	75.376
P19	47.669	1.271	2.648	2.436
P20	60.86	3.148	1.784	1.574
P27	7.039	75.569	66.977	77.536
P30	55.179	62.974	80.205	38.462
P31	60.304	0.000	5.098	0.325
P36	78.663	77.952	77.383	73.115
P37	90.994	87.06	85.197	83.54
P39	28.924	0.112	0.000	0.112
P47	65.401	49.349	46.095	70.282
P101	46.408	10.489	0.000	0.144
P103	96.418	97.851	97.032	97.544
P106	9.708	0.000	0.731	0.000
P108	49.347	4.178	1.828	1.828
P127	61.863	12.518	18.923	54.876
P131	54.484	16.118	21.566	10.897
P136	44.146	2.578	49.517	1.504
P138	69.457	64.341	22.946	68.372
P140	98.732	97.04	97.463	86.047
P159	51.603	58.842	57.601	56.877
P176	94.297	90.326	66.802	90.428
P178	85.98	55.405	61.318	56.588
P190	16.281	18.593	16.181	15.276
P264	31.469	20.979	2.331	20.979
P276	65.589	61.522	59.541	63.295
P279	68.983	16.805	22.925	16.286
P361	47.961	5.579	26.502	20.815
P364	83.294	0.0000	81.776	86.215
P407	85.861	78.449	76.625	87.229
P413	36.975	0.000	0.000	0.000
P449	82.86	49.716	59.478	5.108
P463	83.111	65.778	56.444	56.889
P495	59.406	48.075	35.424	36.634
P527	43.443	36.783	35.246	28.689
P530	22.189	20.382	23.594	23.092
P740	26.496	3.526	6.838	1.603
P937	61.95	0.314	1.887	0.000
P1001	84.736	83.88	58.345	82.026
P1303	36.986	0.000	0.000	0.000
P1376	91.453	91.453	91.026	91.026
P1412	89.99	86.378	81.837	86.171

Table 25: Sentence Typology Relation Comparison BERT-base-cased k@10 accuracy

Sentence Type	Number of Top Positions
Simple	29
Compound	4
Complex	3
Compound-Complex	5

Table 26: Relation Comparison Ranking for k@10 Accuracy in Percent of BERT-Large-Cased on the T-REx Data Set

3. complex: <subject label> has a native language which is <object label>.,
4. compound-complex: <subject label> speaks <object label> and he speaks it well because it is his native language.

We find that, with two active and two passive sentences constructed for this relation, there is no obvious explanation for this result. The relation P27, with label country of citizenship and cardinality N:M, is an obvious outlier because it is the only relation where the other sentence types outperform the simple prompt by a factor of ten. The templates for this relation are as follows

1. simple: <subject label> is <object label> citizen.,
2. compound: <object label> is a country and <subject label> is a citizen of it.,
3. complex: <object label> is the country of which <subject label> is a citizen.,
4. compound-complex: <object label> is a country, with many citizens, for example <subject label> is one of them.

At first glance this seems due to a mistake in the documentation of the LAMA probe because the templates for the simple sentences were all manually optimized prompts taking from the publication, and it is easy to conclude that the template "SUBJ_LABEL is OBJ_LABEL citizen" contains an obvious error in the form of missing the article 'a'. When in investigating this seeming error, however, we find that this is a genuine outlier because adding the "a" results in an equally low result. Another interesting aspect regarding is relation and its accuracy results is the fact that a common sense based assessment of language complexity would tell us that the compound-complex sentence is not only structurally but also generally more complicated to understand, certainly much more complicated than the simple sentence in this case. Relation P30, the continent relation with cardinality 1:N is another outlier because the average accuracy result for the simple sentence type prompt at 55.179% is more than 25 percentage points apart from the top result for the complex sentence type at 80.205%. Overall, the relation comparison does not yield any great new structural insights into the results for BERT-large-cased. Except for a couple of outliers, the simple sentence type dominates the k@10 accuracy performance. The simple sentence prompts reach the highest k@10 accuracy values or are very close to the high score for almost all relations. Therefore, the breaking down of the results into the individual relations has not yielded further insight into the origin of the performance differences between the sentence types or the dominance of the simple results.

4.2.3.3.2 BERT-Base-Cased

The Tables 27 and 28 show the average accuracy results for the four sentence types for each of the cardinality relations for k@1 and k@10 for BERT-base-cased. They strongly resemble the results for the larger BERT-based model. As expected, both results show that the simple sentence type has the highest accuracy for all three cardinalities, again. Once more, the level of accuracy results decreases as the cardinality increases in both sets of results.

Results	Simple	Compound	Complex	Compound-Complex
Total	28.46	13.61	13.02	16.49
1:1	65.10	64.46	63.50	48.13
N:1	32.42	15.17	16.66	20.27
N:M	18.85	7.01	2.87	7.56

Table 27: k@1 Accuracy in Percent of BERT-Base-Cased on the T-REx Data Set

Results	Simple	Compound	Complex	Compound-Complex
Total	57.83	38.87	37.57	39.62
1:1	81.86	81.32	80.79	77.59
N:1	64.38	39.58	44.42	41.33
N:M	44.73	34.38	22.47	33.73

Table 28: k@10 Accuracy in Percent of BERT-Base-Cased on the T-REx Data Set

Once again the simple sentence type based prompts deliver the best average accuracy values across the board, but from these results no clear hierarchy can be established. We will now look at the average k@10 accuracy results for the individual relations. These are demonstrated in Table 29

Here too, we establish a hierarchy based on ranking the different sentence types by the count of top positions in Table 29. Table 30 shows this ranking, and it is identical to the ranking based on BERT-large.

Once more, except for a couple outliers, the simple sentence type dominates the k@10 accuracy performance. The simple sentence prompts reach the highest k@10 accuracy values or are very close to the high score for almost all relations. The structure of the results for BERT-base-cased is a mirror image of the results from the larger model, just on a lower average accuracy level. Therefore, the breaking down of the results into the individual relations has not yielded further insight into the origin of the performance differences between the sentence types or the dominance of the simple results for the smaller BERT-based model.

4.2.3.3.3 BERT-Base-Multilingual-Cased

The Tables 31 and 32 resemble the tables shown for the other two BERT-based models

Relation	Simple	Compound	Complex	Compound-complex
Total	57.83	38.87	37.57	39.62
P17	57.83	38.87	37.57	39.62
P19	47.669	1.271	2.648	2.436
P20	60.86	3.148	1.784	1.574
P27	7.039	75.569	66.977	77.536
P30	55.179	62.974	80.205	38.462
P31	60.304	0.000	5.098	0.325
P36	78.663	77.952	77.383	73.115
P37	90.994	87.06	85.197	83.54
P39	28.924	0.112	0.000	0.112
P47	65.401	49.349	46.095	70.282
P101	46.408	10.489	0.000	0.144
P103	96.418	97.851	97.032	97.544
P106	9.708	0.000	0.731	0.000
P108	49.347	4.178	1.828	1.828
P127	61.863	12.518	18.923	54.876
P131	54.484	16.118	21.566	10.897
P136	44.146	2.578	49.517	1.504
P138	69.457	64.341	22.946	68.372
P140	98.732	97.04	97.463	86.047
P159	51.603	58.842	57.601	56.877
P176	94.297	90.326	66.802	90.428
P178	85.98	55.405	61.318	56.588
P190	16.281	18.593	16.181	15.276
P264	31.469	20.979	2.331	20.979
P276	65.589	61.522	59.541	63.295
P279	68.983	16.805	22.925	16.286
P361	47.961	5.579	26.502	20.815
P364	83.294	0.000	81.776	86.215
P407	85.861	78.449	76.625	87.229
P413	36.975	0.000	0.000	0.000
P449	82.86	49.716	59.478	5.108
P463	83.111	65.778	56.444	56.889
P495	59.406	48.075	35.424	36.634
P527	43.443	36.783	35.246	28.689
P530	22.189	20.382	23.594	23.092
P740	26.496	3.526	6.838	1.603
P937	61.95	0.314	1.887	0.000
P1001	84.736	83.88	58.345	82.026
P1303	36.986	0.000	0.000	0.000
P1376	91.453	91.453	91.026	91.026
P1412	89.99	86.378	81.837	86.171

Table 29: Sentence Typology Relation Comparison BERT-base-cased k@10 accuracy

Sentence Type	Number of Top Positions
Simple	29
Compound	4
Complex	3
Compound-Complex	5

Table 30: Relation Comparison Ranking for k@10 Accuracy in Percent of BERT-Base-Cased on the T-REx Data Set

first in the fact that they show a preference for the simple sentence type regardless of relation cardinality and second in the fact that the average accuracy decreases for four sentence types as the cardinality increases, a phenomenon explained by the increased difficulty level associated with higher cardinalities.

Results	Simple	Compound	Complex	Compound-Complex
Total	19.99	13.40	13.388	10.97
1:1	47.39	39.70	46.74	19.85
N:1	23.35	15.92	16.41	14.19
N:M	18.85	7.05	5.61	4.75

Table 31: k@1 Accuracy in Percent of BERT-Base-Multilingual-Cased on the T-REx Data Set

Results	Simple	Compound	Complex	Compound-Complex
Total	39.29	33.36	30.68	32.17
1:1	70.97	70.44	73.43	65.96
N:1	42.79	35.07	34.81	36.39
N:M	30.85	27.56	20.29	22.31

Table 32: k@10 Accuracy in Percent of BERT-Base-Multilingual-Cased on the T-REx Data Set

Once again it is not possible to establish a clear hierarchy of the other sentence types based on the cardinality results because the non-simple sentence types out and under-perform relative to each other seemingly a random. We will now look at the average k@10 accuracy results for the individual relations. These are demonstrated in Table 33.

Relation	Simple	Compound	Complex	Compound-complex
Total	39.29	33.36	30.68	32.17
P17	43.333	55.161	68.495	55.376
P19	23.199	2.648	4.237	0.53
P20	26.548	1.784	3.987	4.827
P27	0.311	54.141	54.555	62.112
P30	10.359	94.154	79.077	77.744
P31	48.698	0.217	0.759	2.603
P36	67.141	69.417	69.417	60.882
P37	83.747	48.033	54.037	72.36
P39	10.65	0.112	0.112	0.112
P47	41.54	46.312	40.239	54.772
P101	19.684	5.316	0.718	1.724
P103	93.04	96.315	91.095	88.946
P106	27.975	0.104	2.923	0.104
P108	4.7	1.305	0.000	1.305
P127	25.91	5.24	11.645	43.814
P131	32.804	19.75	18.275	13.167
P136	22.342	1.826	17.83	3.759
P138	35.969	43.876	15.814	47.597
P140	82.875	63.214	78.436	0.000
P159	50.569	52.327	49.535	50.052
P176	90.428	88.086	50.916	87.576
P178	49.662	62.331	57.432	58.784
P190	12.965	11.357	13.367	12.764
P264	4.429	1.632	0.233	0.932
P276	51.408	56.309	56.517	56.1
P279	48.133	33.714	15.145	22.095
P361	34.227	25.107	20.386	20.172
P364	78.505	8.645	73.481	65.537
P407	52.794	73.318	68.529	76.397
P413	2.311	0.000	0.000	0.000
P449	73.893	4.427	4.313	2.951
P463	68.444	46.222	4.444	37.778
P495	2.09	42.354	35.974	40.044
P527	33.504	29.508	19.775	24.795
P530	17.871	14.759	25	8.735
P740	1.389	3.526	4.915	3.526
P937	56.709	1.782	0.419	0.839
P1001	82.454	80.884	72.325	40.086
P1303	13.804	0.000	0.527	0.000
P1376	82.479	73.504	85.47	81.197
P1412	53.664	78.019	10.01	42.724

Table 33: Sentence Typology Relation Comparison BERT-base-multilingual-cased k@10 accuracy

For the multilingual model, it seems a much more fruitful endeavor to establish a hierarchy between all four relations based on the individual relation results shown in Table 33 because at first glance, there seem to be more top positions for the non-simple sentence types. Table 34 shows the ranking of the sentence types according to the number of top positions. It differs from the rankings of the other two BERT-based models in the fact that no the compound-complex sentence type ranks in second place but the last place because compound and complex sentence types share place two. Interestingly, the compound-complex sentence type has the same amount of top positions as for the other BERT-based models, but the relations are neither congruent nor disjoint.

Sentence Type	Number of Top Positions
Simple	22
Compound	7
Complex	7
Compound-Complex	5

Table 34: Relation Comparison Ranking for k@10 Accuracy in Percent of BERT-Base-Cased on the T-REx Data Set

The multilingual BERT-based model again has most outliers in common with the other BERT-based models. Once again, there is no clear pattern within the individual relations results to explain the dominance in fact retrieval performance of the simple sentence type or the general differences between the sentence types, leading to the conclusion that these differences could stem from the way the model processes language itself.

4.2.3.4 Score Comparison

Up until this point in the investigation we have only examined different accuracy values in order to assess, understand and analyze the differences between the different sentence types. But these accuracy measures are ultimately just ranking based results because the models return probability scores for all tokens. It is therefore natural to extend this analysis to the probabilities the models assign to the token belonging to correct object for a single triplet, the so-called correct token probability. In this analysis, we will take a look at the scores and compare them for the different sentence types and cardinalities and construct a probability distribution for each of those and compare these using the KL-Divergence.

4.2.3.4.1 BERT-Large-Cased

Table 35 shows the average correct token scores for all four different sentence types. And as it is inline with the various accuracy results discussed above, we find the exact same pattern as before. The simple sentence type has the highest average correct token scores, outperforming the other sentence types by a considerable margin. The differences in scores among the non-simple sentence types are significantly lower than the differences

compared to the simple sentence type.

Name	Simple	Compound	Complex	Compound-complex
Total	0.035	0.02	0.017	0.022
1:1	0.13	0.129	0.122	0.089
N:1	0.042	0.022	0.02	0.025
N:M	0.015	0.009	0.004	0.011

Table 35: Sentence Typology Correct Token Scores BERT-large-cased

Moving on from the total result to the cardinalities we observe the same phenomenon, as observed in the relation comparison in the previous paragraph, the correct token scores are highest for the simple sentence types' prompts and the all scores decrease as the cardinality increases. In order to gain a better understanding of the correct token scores, we analyze their distribution by constructing a probability distribution counting the number of occurrences within 99 bins, consecutive intervals placed between 0 and 1. Figure 10 shows a plot mapping the correct token scores and their respective probabilities.

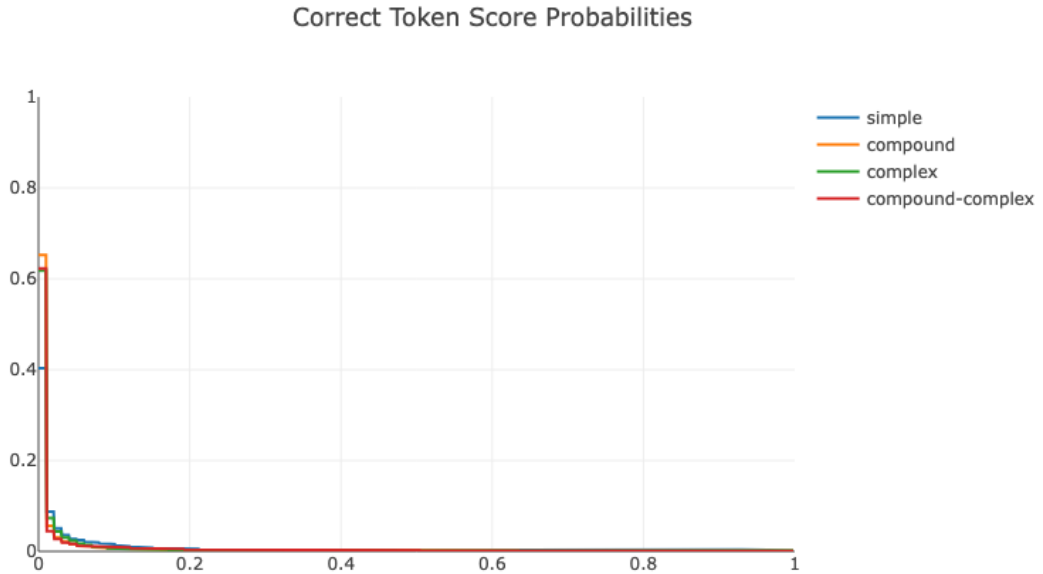


Figure 8: Sentence Typology Score Probability Distribution BERT-large-cased

Obviously, Figure 10 shows that most of the correct tokens scores are clustered around the origin and the more the score values approach the value 1 the closer to 0 the probabilities get. At this angle the following dictum holds, the as the correct token score increases its probability decreases. This holds for all four sentence types. However, if zoom into the plot in order to better understand what is going on at the top of the correct token scores, we can see more clearly.

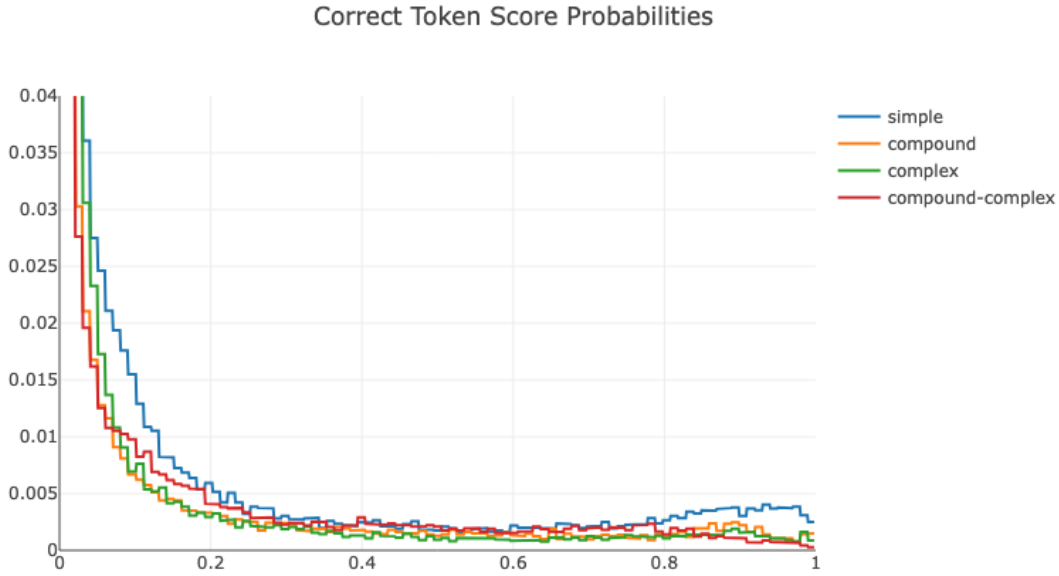


Figure 9: Sentence Typology Zoomed Score Probability Distribution BERT-large-cased

The plot in Figure 9 enables the differences between the sentence types' different probability distributions to appear much more visible. At first, we observe that the curves for the complex and the compound sentence type are very similar to each other, whereas the curves for the simple sentence type and the compound-complex sentence really do differ from them and from each other. These differences for both of the differing types are found at the end of the probability scales. Figure 11 shows that the simple sentence starts from a much lower global maximum of 40% probability at global maximum at zero as compared to the other sentence types global maximum around 60%. Furthermore, the simple sentence type begins its descent from the global maximum much later than the other curves and has a less steep descent overall. This means more of the probability mass is located further from zero compared to all the non-simple sentence types. Translated in terms of fact retrieval performance, this denotes that the model generally assigns more probability to correct tokens than for all three other models, ultimately meaning the correct token is generally more likely to be predicted by the model than for the others. The second aspect is the fact that the plot for the simple sentence type reaches a second local maximum around the end of the axis. This indicates conclusively that higher score values are much more likely to be assigned to the prompts generated using the simple sentence templates than the others. Thus, the bulk of probability mass is assigned to values smaller than twenty and larger than 80%. This is inline with our findings so far because it is an explanation for the general dominance of the simple sentence type in almost all slices of the data, aside from some particular relations. A less steep descent is also the one characteristic that separates the compound-complex probability plot from the plots of the complex and compound sentence types. It assigns more probability mass at the low end of the scale than the other two models but also less at

the other, the high end, of the spectrum. The next tool we employ to understand the differences between the sentence types by analyzing the scores and their probabilities is by comparing the different distributions directly using the KL-Divergence, which measures the difference of two probability distributions by calculating how good of a substitute one distribution is for another one. Figure 10 shows a heat map of the KL-Divergence calculated for any pair of the four different sentence types. The very clear result of this analysis is that the fact that the simple sentence type once again emerges as the type having most different probability distribution from all other types. This is demonstrated by the big blue block in the right corner of the heat map. Furthermore, and again, the compound-complex sentence type appears to be the second most different sentence type compared to all others based on the correct token score probability distribution. Combined, these measures all show that there is a clear difference in the models responses to the four sentence types. This difference is once again most pronounced for the simple and second most pronounced for the compound-complex sentence type.

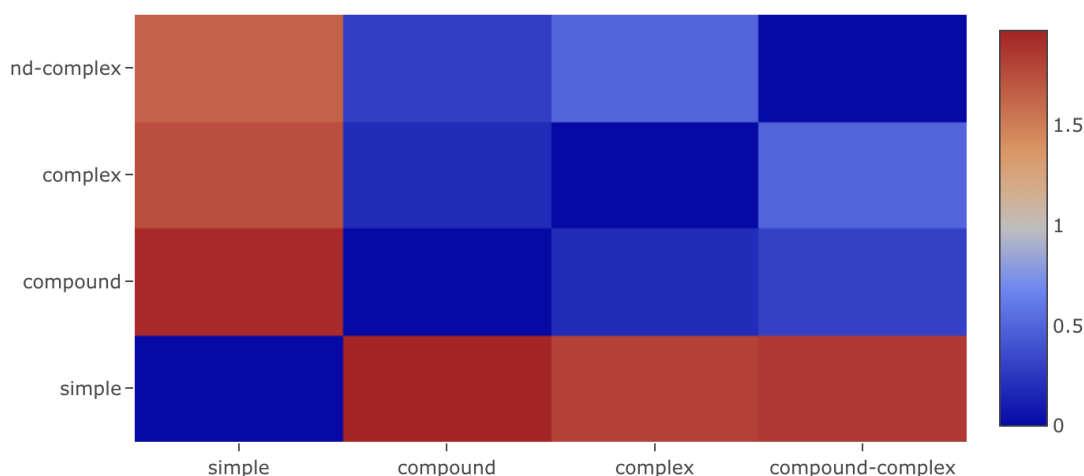


Figure 10: Sentence Typology KL-Divergence BERT-large-cased

4.2.3.4.2 BERT-Base-Cased

We will now take a closer look at the scores and their probability distribution for the smaller of the BERT-based models. Table 36 shows the correct token probability for the BERT-base-cased model, and the table paints almost the mirror image of its larger version. The correct token score is at the highest value for the simple and at the smallest for the complex sentence type. This pattern holds across all three different cardinalities and as before the correct token score decreases as the cardinality increases because of a higher level of difficulty.

Name	Simple	Compound	Complex	Compound-complex
Total	0.035	0.02	0.017	0.022
1:1	0.13	0.129	0.122	0.089
N:1	0.042	0.022	0.02	0.025
N:M	0.015	0.009	0.004	0.011

Table 36: Sentence Typology Correct Token Scores BERT-base-cased

This is also true for the plot of the probability distributions of the correct token scores for all four sentence types for BERT-base-cased shown in 11 which is the mirror image of the distribution shown for BERT-large-cased.

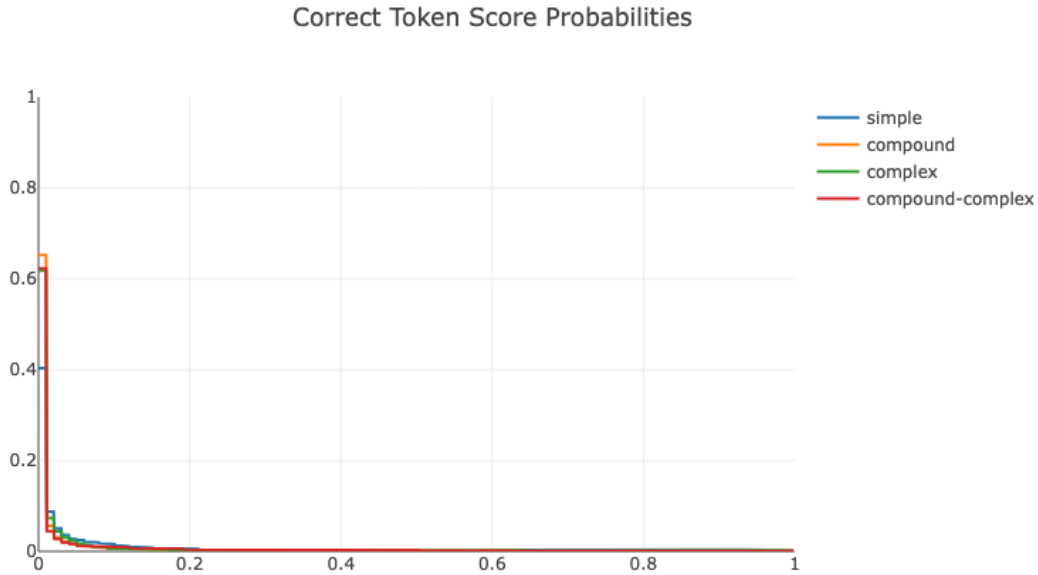


Figure 11: Sentence Typology Score Probability Distribution BERT-base-cased

Furthermore, the effect analyzed for BERT-large cased at the ends of the scale of the plot axes, namely the differences at the extremes among the different distributions belonging to the four sentence types, is present for BERT-base-cased also. Once again, the simple sentence type's correct token score probability distribution is most different from the others, especially at the beginning and the end of the axis. The compound-complex distribution is also very different from the other distributions, but the complex and the compound probability plots are very similar. The zoomed in plot, see Figure 12 also shows the second local maximum at the end of the scale that was also observable in the Figure 9 for the larger BERT-based model.

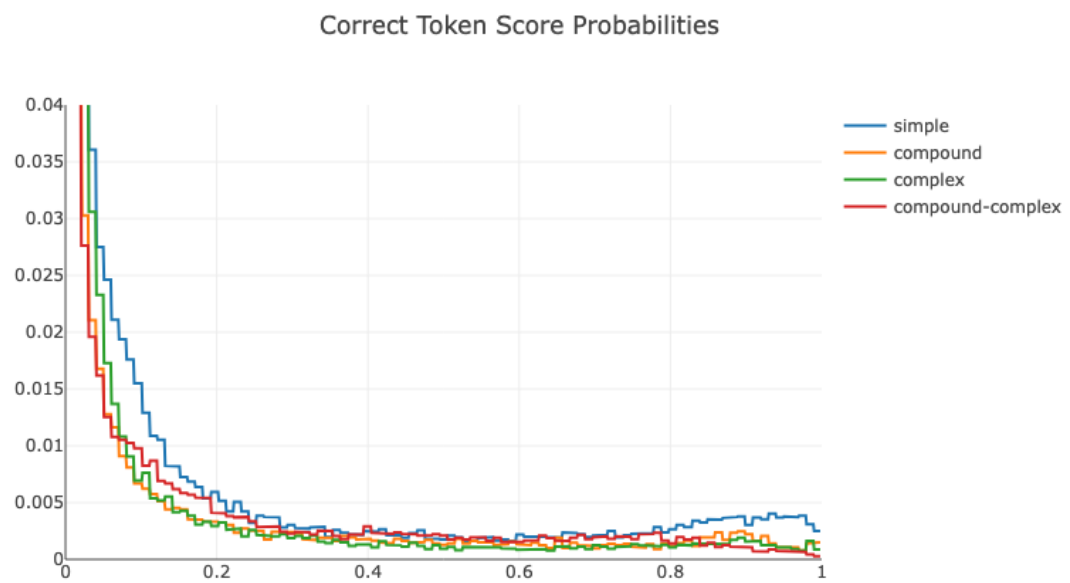


Figure 12: Sentence Typology Zoomed Score Probability Distribution BERT-base-cased

Next we will again compare the probability distributions for the four different sentence types using the KL-Divergence. Figure 13 shows a heat map constructed from this comparison, and it shows the same big blue block in the upper right corner denoting that the three non-simple sentence types are very similar to each other when compared to the distribution of the simple sentence type and all most dissimilar to it with the complex sentence being most similar to the simple sentence type. Among the non-simple sentence types, the compound and the complex sentence type appear to be the best substitutes for each other and therefore the most similar sentence types.

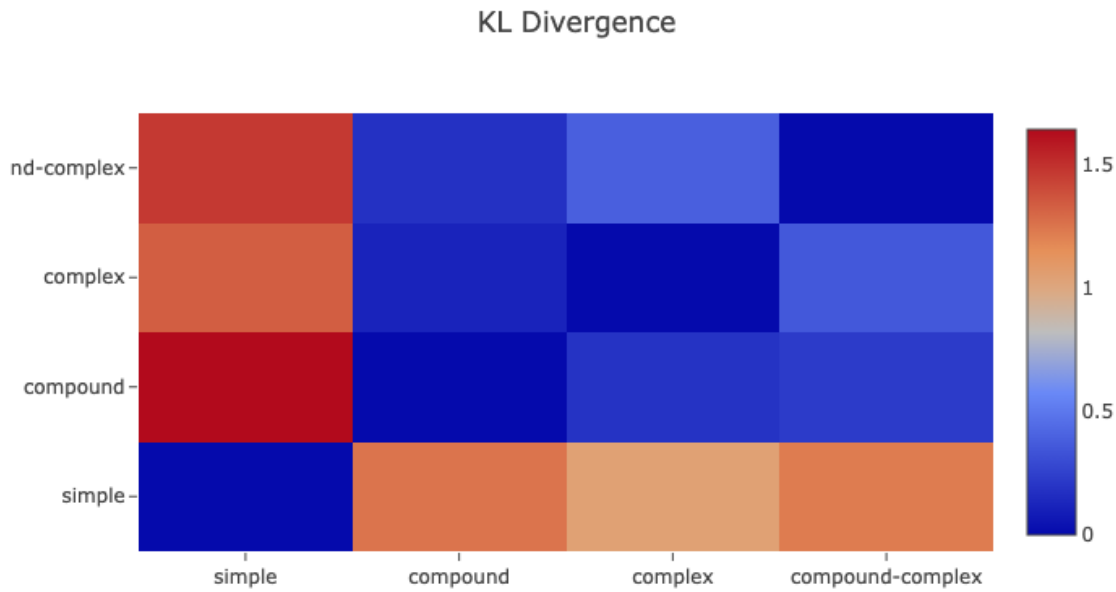


Figure 13: Sentence Typology KL-Divergence BERT-base-cased

4.2.3.4.3 BERT-Base-Multilingual-Cased

Finally, we will take a look at the scores and their probability distribution for the BERT-base-multilingual-cased model. Table 37 shows the average token scores for the correct token for the multilingual BERT-based model. The observations are once again the same, the correct token score is highest for the simple and smallest for the complex sentence type. This pattern holds across all three different cardinalities and, as seen twice before, the correct token score decreases as the cardinality increases because of a higher level of difficulty.

The probability distribution of the correct token scores for the multilingual model, see Table 14, and it's zoomed in version Table 14, demonstrate the same behavior as the distributions already analyzed in detail in both previous analyses. The outliers are the compound-complex and the simple sentence type, which shows the delayed decline,

Name	Simple	Compound	Complex	Compound-complex
Total	0.035	0.02	0.017	0.022
1:1	0.130	0.129	0.122	0.089
N:1	0.042	0.022	0.020	0.025
N:M	0.015	0.009	0.004	0.011

Table 37: Sentence Typology Correct Token Scores BERT-base-multilingual-cased

lower global and now characteristic second local maximum.



Figure 14: Sentence Typology Score Probability Distribution BERT-base-multilingual-cased

Figure 16 shows the direct distribution comparison between the sentence types, as before, using the KL-Divergence. The heat map looks almost identical to the previously shown heat maps for the other two BERT-based model, it again shows the big blue block in the upper right corner denoting the probability similarities between the non-simple sentence types to be greater than compared to the simple sentence type's distribution.

This analysis again has not provided us with a clearer picture of the differences between the sentence types by supplying more details about how and where these differences occur, but has still not yielded an explanation of their origin.

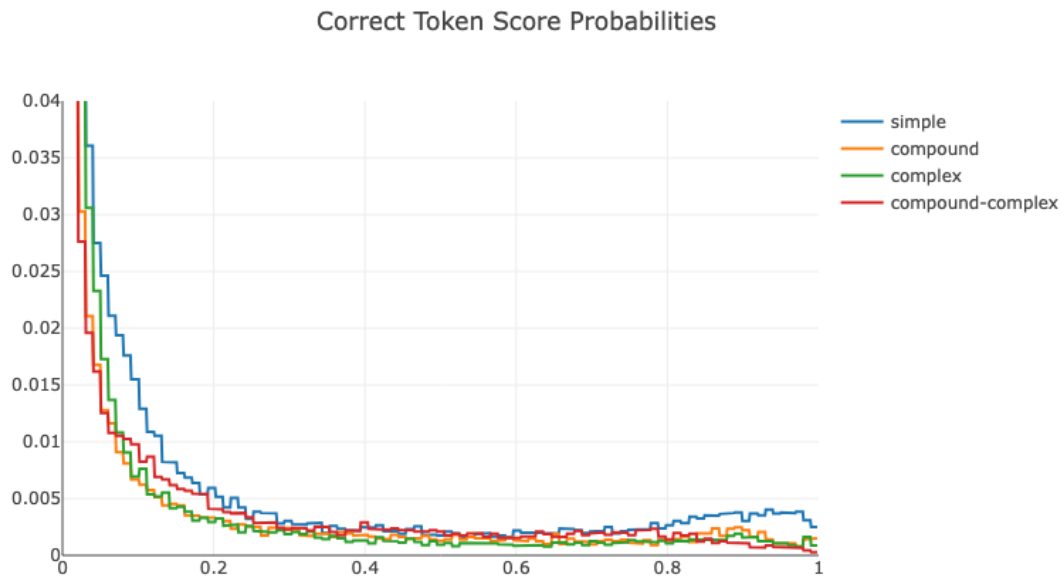


Figure 15: Sentence Typology Zoomed Score Probability Distribution BERT-base-multilingual-cased

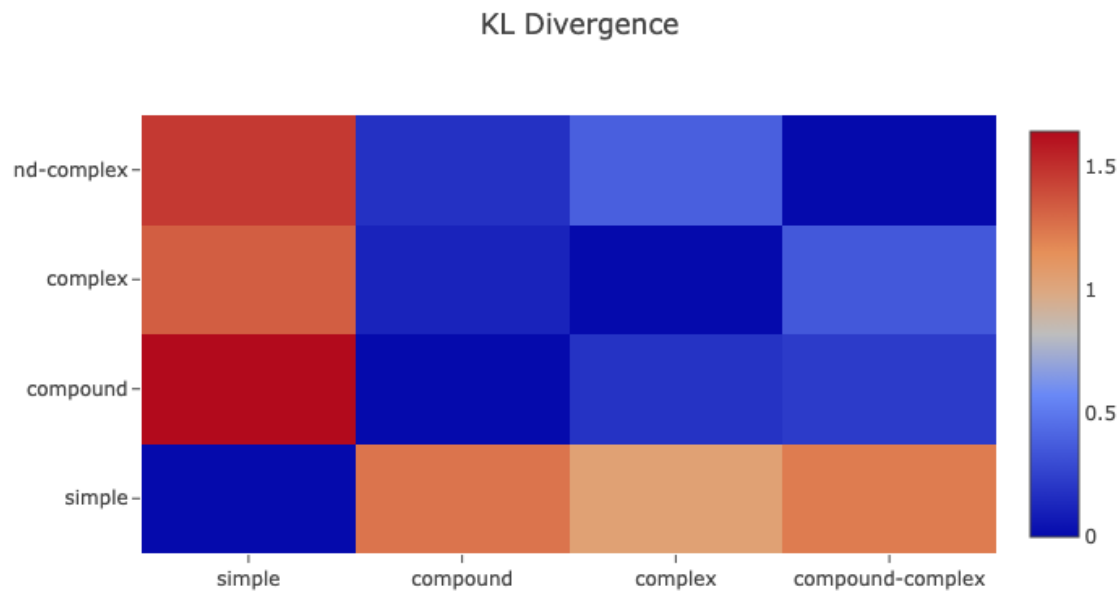


Figure 16: Sentence Typology KL-Divergence BERT-base-multilingual-cased

4.2.3.5 Embedding Comparison

So far we have compared different models according to accuracy, analyzed different accuracy values for the top performing models, analyzed the results for different segments of the data set in detail and analyzed the scores assigned by the models to the correct tokens. One unexplored avenue of investigation remains in the form of the models embeddings. The embeddings are the final representation of the input as processed by the model. They are the final-hidden states and since they are used further as input to downstream tasks in fine-tuning, they are often called features. Thus, these features represent the final interpretation of the input the model arrives at. For this reason, we will analyze and compare these features for the four different sentence types again for the top performing models. In order to be able to compare these features, we will use the metric cosine similarity to again ask how similar the embeddings for the different sentence types are to each other. We will again perform this analyses for the three different top performing models. Figure 17 shows a heat map constructed from the cosine similarities between the embeddings of the different sentence types. The cosine similarity measure is a measure that projects the values into the interval $[-1, 1]$. Thus, the fact the similarity values span only from 0.5 to 1 is already an indication that the embeddings themselves are not completely dissimilar. This is interesting because so far we did not know anything about the inner representation of our four different sentence types, except for the fact if the model is able to predict the correct token. But generally, it would have been possible that the model reached two correct predictions based on two completely dissimilar sets of features. But the model seems to understand in some way that the sentences, which are completely equal in semantics and only differ in syntax, are themselves quite similar. Unsurprisingly the model finds the simple sentence type most dissimilar to the non-simple sentence types, this dissimilarity between the representations seems to be responsible for the large fact retrieval performance gap. Furthermore, compound-complex and compound also lead to highly similar representations with a measure above 0.8. Generally, this heat map shows that all three non-simple sentence types lead to model features which are highly similar.

The heat map for the cosine similarity for BERT-base-cased, Figure 18, shows a picture that is different in some details from its larger version's, for example the higher similarity in representations of the compound and the complex sentence type, but also generally affirms the observations made for BERT-large.

These observations are then further strengthened by the cosine similarity results for BERT-base-multilingual cased, which are shown in Figure 19.

4.2.4 Further Analysis and Interpretation

In this section we will further analyze, interpret and discuss the results of the sentence typology experiments described in the section above in order to find out more about the differences produced by the models in their interpretation of the four different sentence types. Therefore, we undertake further calculations and analyze results using additional metrics in order to explain the superiority of the simple sentence type and other differences between the sentence types. We will once again conclude that there is no simple explanation for these differences that are based on superficial differences between the

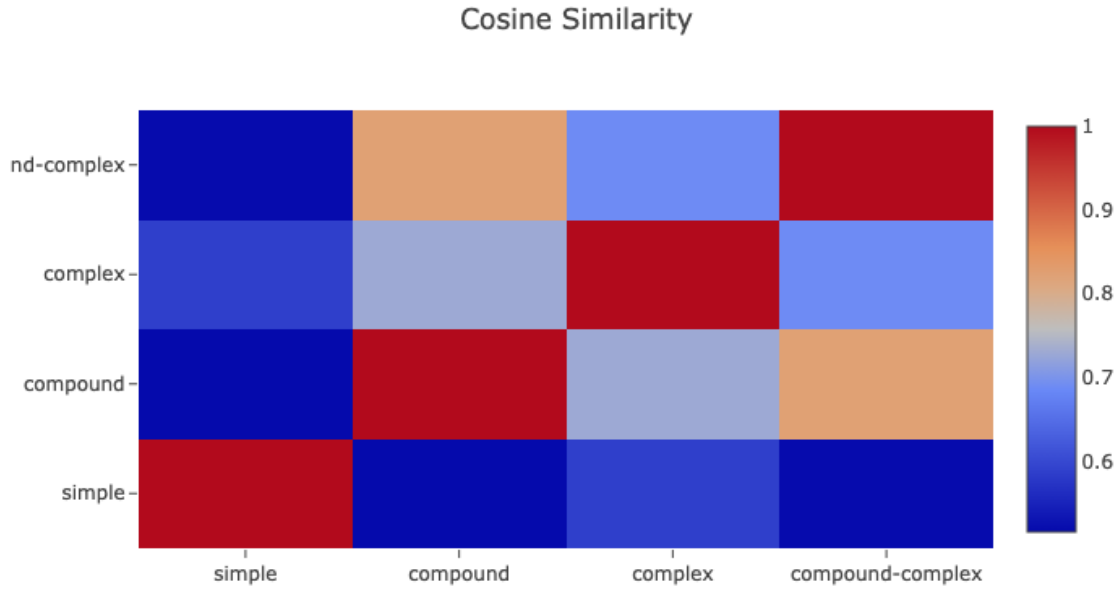


Figure 17: Sentence Typology Cosine Similarity Heat Map for BERT-large-cased

prompts generated for the sentence types.

4.2.4.1 Information Density

A hypothesis one quickly arrives at when studying the fact retrieval accuracy results for the BERT-based models is about information density. Given the fact that the semantic content of each sentence type was roughly equal for all relations but the non-simple sentence types expressed that semantic content in a greater amount of words, it is natural to pose the question if the ratio between information and sentence length could be an explanation for the performance differences in fact retrieval accuracy. Table 39 shows the length of the template for each sentence type for all relations, and Table 38 shows the average length for the four types of sentences. Table 38 shows that the simple sentences are indeed the shortest, whereas the compound-complex sentences are the longest.

Sentence Type	Length
simple	9.59
compound	14.93
complex	14.15
compound-complex	20.07

Table 38: Average Length for the sentence typology for the T-REx Data Set

In order to investigate the influence of the information density, we analyze the correlation

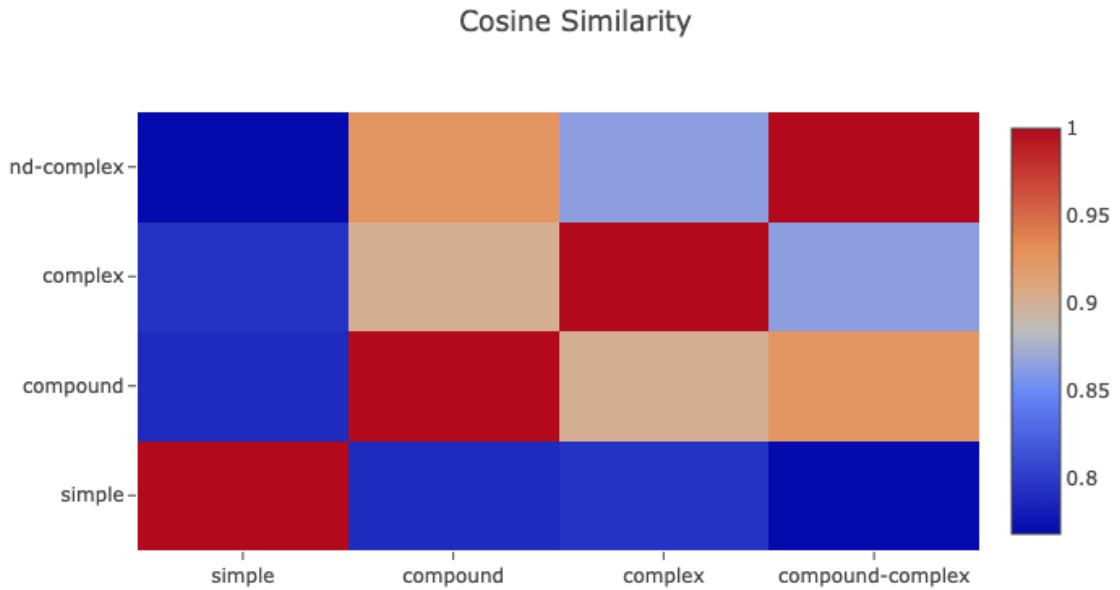


Figure 18: Sentence Typology Cosine Similarity Heat Map for BERT-base-cased

between the sentence length and the $k@1$ and $k@10$ accuracy levels, as well as the probability of the correct token. The length was measured as the number of tokens extracted from the template by the tokenizer.

We can see from Table 40 that the correlation between length and accuracy or correct token probability is exceedingly weak. In order to explain the dominance of the simple sentence type with the information density, we would need to see a strong negative correlation between the accuracy and sentence length. Perhaps since simple sentences achieve the highest accuracy levels overall and are the shortest sentences, the correlation would be stronger if we only considered the data for the simple sentences and one of the other sentence types and calculate the accuracy. But as Table 41 shows, these correlations do also not yield any conclusive correlations.

4.2.4.2 Subject and Object Position

Another possible explanation for the model's higher accuracy levels for simple sentences compared to other sentence types could be the position of the subject and object labels. There are two possible avenues to explore. Perhaps the order of the subject and the object label is relevant to the prediction. Furthermore, since simple sentences tend to be shorter, but length did not explain a large part of the variance in the results, the distance between the subject and the object label is the right measure to consider. In order to investigate the influence of the order, we simply determine the order of the subject and the object label for every template for all relation. If the order is the standard order, subject followed by

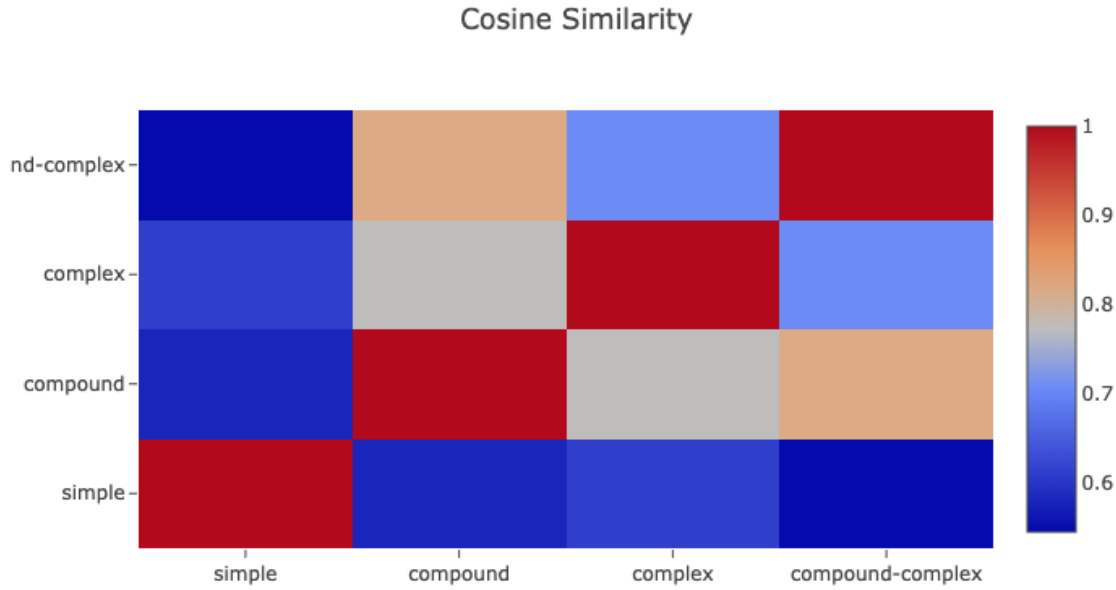


Figure 19: Sentence Typology Cosine Similarity Heat Map for BERT-base-multilingual-cased

the object, we assign the value 1 and if the order is reversed we assign the value of 0. With these new data points, we can calculate the correlation between the order and the k@1 accuracy. Table 42 shows the results of this calculation.

To calculate the influence of the distance between the subject and the object labels of a sentence, we must first define what we mean by distance. We could just measure the distance as the amount of characters or words in the string between the end of one label and the beginning of the other. But since the tokenizer of the model does more than just split the sentence into words, it seems to be more adequate to measure the distance between the labels' tokens. Table 43 shows the results of this calculation.

As we can see from these results the correlation between our distance metric and the accuracy and correct token probability is very weak but the correlation between order and probability is the strongest correlation we have seen so far.

4.2.4.3 Reducing the Data

In order to understand the data why the accuracy levels are higher for simple sentences than the others, we can also concentrate our investigation on the relations for which we can even say that the simple sentences result in a reasonable good accuracy level in retrieving facts. Thus, we reduce the data to relations for which the accuracy for BERT-Base is at least 50% because this way the model correctly predicts the object for more triples

Relation	Simple	Compound	Complex	Compound-Complex
P1001	11	13	15	20
P101	11	18	19	23
P103	11	14	13	18
P106	10	18	12	27
P108	8	14	12	18
P127	9	13	11	19
P1303	7	15	17	21
P131	9	14	13	18
P136	8	13	15	23
P1376	10	16	14	19
P138	9	17	16	22
P140	12	18	17	28
P1412	10	15	15	21
P159	13	16	15	20
P17	9	13	14	19
P176	9	15	12	19
P178	9	17	11	22
P19	9	16	16	23
P190	10	16	19	23
P20	8	15	15	20
P264	11	13	13	21
P27	8	15	14	20
P276	9	14	13	18
P279	12	17	16	18
P30	9	13	13	18
P31	8	16	14	19
P36	10	15	15	19
P361	9	15	13	18
P364	11	16	15	22
P37	11	16	15	20
P39	10	14	13	18
P407	9	15	15	22
P413	9	13	12	20
P449	10	13	13	21
P463	10	15	14	17
P47	9	15	16	16
P495	9	14	13	18
P527	8	13	13	18
P530	10	17	14	21
P740	9	13	11	17
P937	10	14	14	19

Table 39: Length of the relation for the sentence typology for the T-REx Data Set

Indicator	Model	Pearson's r	Spearman's Correlation
k@1 Accuracy	BERT-Base-Cased	-0.114	-0.171
k@10 Accuracy	BERT-Base-Cased	-0.143	-0.146
Probability	BERT-Base-Cased	-0.136	-0.185

Table 40: Correlation between Length and Indicator for the Sentence Typology for the T-REx Data Set

k@	Types	Pearson's r	Spearman's Correlation
1	Simple and Compound	-0.2146	-0.1594
1	Simple and Complex	-0.2054	-0.1792
1	Simple and Compound-Complex	-0.2554	-0.1588
10	Simple and Compound	-0.1540	-0.2250
10	Simple and Complex	-0.1470	-0.2013
10	Simple and Compound-Complex	-0.2163	-0.2367

Table 41: Correlation between Length and Indicator for the Sentence Typology for the T-REx Data Set

Indicator	Model	Pearson's r	Spearman's Correlation
k@1 Accuracy	BERT-Base-Cased	0.193	0.253
k@10 Accuracy	BERT-Base-Cased	0.235	0.251
Probability	BERT-Base-Cased	0.304	0.360

Table 42: Correlation between Subject-Object-Order and Indicator for the Sentence Typology for the T-REx Data Set

Indicator	Model	Pearson's r	Spearman's Correlation
k@1 Accuracy	BERT-Base-Cased	-0.132	-0.117
k@10 Accuracy	BERT-Base-Cased	-0.063	-0.54
Probability	BERT-Base-Cased	-0.088	-0.227

Table 43: Correlation between Subject-Object-Distance and Indicator for the Sentence Typology for the T-REx Data Set

than it misses. This seems to be a good standard for a reasonably well accuracy level. The relations for which this condition holds are the following, shown in Table 44.

The percentage in the fifth column of Table 44 shows that the average accuracy as a percentage of the simple accuracy differs widely and ranges from 2% to more than 100%. Concentration on only on these relations, we can apply our previously performed analyses again.

Table 45 shows that the correlation between the correct token probability and the order of the sentence is comparatively high and with a value of up to 5.7% can explain more than 30% of the variance in the data. This means that in these cases, where the accuracy levels of a relation is above 50% is tied to the position of the mask in the sentence. Positioning

Relation	Name	Simple	Others avg.	Percentage of Simple
P1412	languages spoken, written or signed	65.0	1.3	2.0%
P463	member of	67.1	3.1	4.6%
P178	developer	58.6	22.2	37.9%
P407	language of work or name	59.2	25.7	43.4%
P37	official language	54.6	35.2	64.6%
P140	religion or worldview	75.1	53.3	71.1%
P176	manufacturer	85.6	62.5	72.9%
P36	capital	62.2	53.8	86.6%
P1376	capital of	73.9	73.4	99.2%
P103	native language	72.2	74.1	102.7%

Table 44: Limit for Simple @50% k@1 Accuracy in Percent of Simple Accuracy of data the T-REx Data Set

Performance Indicator	Property	Pearson's r	Spearman's Cor.
k@1 Accuracy	Length of Sentence	-0.009	-0.027
k@1 Accuracy	Distance of Sub, Obj	-0.130	-0.119
k@1 Accuracy	Order of Sub,Obj	-0.020	-0.045
k@1 Accuracy	starts with Sub	-0.117	-0.120
k@1 Accuracy	starts with Mask	-0.019	-0.045
Correct Token Prob.	Length of Sentence	-0.207	-0.238
Correct Token Prob.	Distance of Sub, Obj	-0.116	-0.157
Correct Token Prob.	Order of Sub,Obj	0.571	0.550
Correct Token Prob.	starts with Sub	0.502	0.489
Correct Token Prob.	starts with Mask	0.571	0.550

Table 45: Correlation between Performance Indicator and Property for BERT-Base-Cased for the Sentence Typology for the T-REx Data Set

the mask at the beginning of the sentence seems to impede the performance drastically.

4.2.4.4 Cased vs. Uncased

One result we can state unequivocally is the fact that the most important aspect regarding the prediction accuracy in fact retrieval task seems to be the vocabulary. In the English language People, Countries, Companies and other named entities are capitalized and therefore many important pieces of information one might want to retrieve from a knowledge base are related to these named entities and are dependent on this capitalization. The results of this investigation suggest that the vocabulary needs to reflect that. For a fact retrieval tasks, it makes a huge difference if we want to retrieve a fact about Apple the company or about an apple that is a fruit. This ambiguity can be mitigated by working with a cased vocabulary, which is very much reflected in the results of the experiments.

4.2.5 Abstract vs Real World Entities

In our experimentation with the T-REx data source, we are always confronted with three different kinds of triples. Those that exclusively deal with real world entities, we will call them pure real world triples. A pure real word triple is a triple where both its subject and its object refer to real world entities such as a country, city, person, company product, etc. Two examples of a pure real world triple comes from the relation P36 with the triple (Azerbaijan, capital, Baku) and from the relation P178 with the triple (MessagePad, developer, Apple). In the second category we have those triples that exclusively relate abstract entities to each other via relation, we will call them pure abstract triples. An abstract entity is an entity that is itself a category of things more than it is a thing itself. Two examples of pure abstract entities are from relation P361, (arithmetic, part of, mathematics) and from relation P279 (white bread, subclass of, bread). In the last category we have mixed triplets where the object is an abstract entity and the subject is a real word entity or vice versa. Two examples of mixed triples are from relation P31 (Tantallon Castle, instance of, castle) and from relation P39 (Bernard Francis Law, position held, archbishop). Although a PLM that is supposed to be used as a knowledge base should work equally well for abstract and real world entities, there could be difference in the way the model processes them which could give us a new insight into the accuracy differences for the sentence typology. In order to investigate this, we first identify the pure abstract tokens and construct a partial T-REx data source consisting of them. Then we perform the same experiments used above for this restricted data source.

4.2.5.1 Abstract T-REx Data Source

In an effort to identify the relations that contain pure abstract triples and then identifying the pure triples themselves, we need a heuristic to automatically decide if a subject or object label represents an abstract or a real world entity. Fortunately, the English language makes this task rather easy through the rule that names, which are overwhelmingly reserved for real word entities, are capitalized, and all other subject are not capitalized. Therefore, we can roughly identify all pure abstract triples by limiting all triples to those whose subject and object label are not capitalized. Unfortunately, this leaves us with a rather small percentage of the overall data source with just 1.715 triples, the breakdown is shown in Table 46. As we can see, the vast majority of the pure abstract triplets is concentrated in only three relations that very naturally fit the conditions.

4.2.5.2 Results

The restriction to these pure abstract triples also has the additional benefit that we should not see any difference between cased and uncased versions of models because all triples with capitalized subject and object labels have been excluded. Therefore, we begin this analysis with a model comparison, comparing the results once again for the four different sentence types from the sentence typology and for all six BERT-based model, before we describe and analyze the results for both BERT-Base models on the abstract triples.

Relation	Name	Cardinality	Number of triples
P31	instance of	N:1	32
P101	field of work	N:M	27
P138	named after	N:1	16
P276	location	N:1	3
P279	subclass of	N:1	532
P361	part of	N:1	470
P527	has part or parts	N:M	634
P937	work location	N:M	1
Total	-	N:1/N:M	1.715

Table 46: Pure Abstract Triples of the T-REx Data Source

4.2.5.2.1 Model Comparison

Table 47 shows the average k@10 accuracy results for these models on the sentence typology data set created from the abstract T-REx data source. These results indicate that the huge differences in accuracy levels between the cased and the uncased models are indeed caused by the real world entities contained in the T-REx data source because the differences between the cased and the uncased models shrink significantly. In the case of both BERT-base-multilingual models, the performance hierarchy even turned around completely, meaning the uncased version outperforms the cased model by a notable margin.

Model	Simple	Compound	Complex	Com-com
BERT-large (c)	61.924	33.528	32.303	27.464
BERT-large (uc)	53.761	45.714	25.539	47.172
BERT-base (c)	59.475	23.965	31.545	17.959
BERT-base (uc)	51.254	40.933	34.519	48.047
BERT-base-multil. (c)	40.117	32.245	19.125	20.816
BERT-base-multil. (uc)	45.19	32.536	33.703	37.085

Table 47: Sentence Typology Model Comparison abstract k@10 accuracy

4.2.5.2.2 BERT-base Comparison

Once again, the results, see Tables 48 and 49, show that the models predict the abstract objects with the highest k@1 accuracy for the simple sentences. As Table ?? demonstrates, this is especially true for the three large abstract relations, P279, P361 and P527 except for relation P527 for the uncased version of BERT-Base. Another interesting observation is that for the abstract entities the uncased version of BERT-base-multilingual outperforms the cased version, indicating that although the uncased model contains almost no information about real world entities it seems to have more knowledge about abstract entities.

The explanatory power of these results and interpretations is severely hindered by the

Result	Type	Simple	Compound	Complex	Compound-Complex
Total	Cased	19.36	4.78	7.81	2.74
Total	Uncased	24.61	9.97	12.13	11.25
N:1	Cased	22.22	0.76	7.41	0.09
N:1	Uncased	32.86	10.07	12.35	9.31
N:M	Cased	14.80	11.18	8.46	6.95
N:M	Uncased	11.48	9.82	11.78	14.35

Table 48: k@1 Accuracy in Percent of BERT-Base-Cased on the Abstract T-REx Data Set

Result	Type	Simple	Compound	Complex	Compound-Complex
Total	Cased	59.48	23.97	31.55	17.96
Total	Uncased	51.25	40.93	34.52	48.05
N:1	Cased	62.20	11.97	27.54	10.35
N:1	Uncased	62.68	36.09	34.38	52.99
N:M	Cased	55.14	43.05	37.92	30.06
N:M	Uncased	33.08	48.64	34.74	40.18

Table 49: k@10 Accuracy in Percent of BERT-Base-Cased on the Abstract T-REx Data Set

small amount of triples and the kind of relations contained in the abstract T-REx dataset.

4.2.5.3 Syntax Dependency Tree Metrics

In this section, we once try to explain the variance in the results between the different sentence types by using structural approaches of the sentences build from the templates. Therefore, we construct syntax dependency trees with the library spacy and try to explain the variance with graph properties of these trees. We try a number of different metrics derived from such tree properties, but none of them can explain the variance in a meaningful way. Table 50 shows the correlation between Shortest-Path-Length and the k@1, the k@10 accuracy and the correct token scores and we can see that this property of the prompts also does not explain the variance.

Indicator	Model	Pearson’s r	Spearman’s Correlation
k@1 Accuracy	BERT-Base-Cased	0.060	0.005
k@10 Accuracy	BERT-Base-Cased	0.040	0.039
Probability	BERT-Base-Cased	0.207	0.242

Table 50: Correlation between Shortest-Path-Length and Indicator for the Sentence Typology for the T-REx Data Set

4.3 Active, Passive and Nominalization

In this section, we explore the impact of syntactical structures on fact retrieval performance of PLMs by comparing their overall accuracy for different verb morphology types, namely sentences with active, passive and nominalized verb morphologies. In this experiment we prompt the PLMs Bert-Base-Cased and others with prompts created from templates of active, passive and nominalized sentences in order to find, analyze and compare differences in the accuracy levels of the answers and among the PLMs. We perform this experiment only on a suitable subset of T-REx triplets. This subset contains the following relations

- P127 - owned by,
- P136 - genre,
- P176 - manufacturer,
- P178 - developer,
- P413 - position played on tea,
- P1303 - instrument.

These six relations are chosen based on the criteria that the respective manually constructed LAMA prompt template for the relation contained a verb in active or passive form. The objective was to be able to construct reasonable templates for the active, passive and nominalized sentences.

4.3.1 Templates

For each of the thus chose six relations of the T-REx data source we create a set of three templates, one sentence template for each of the verb categories active, passive and nominalized. The sentence templates were manually created with the intent of carrying roughly the same semantic content or information as far as that was possible. It would be much easier to construct sentence type templates without this restriction and the sentences would often sound more natural but it was important to keep the semantic content equivalent as much as possible in order to really observe the effect of the changing syntactical structure. The following two examples give an impression of these templates. The full set of templates can be viewed in the Appendix.

4.3.2 Experimental Setup

In this experiment, the PLMs under investigation are queried with three sentences each for all triples of the specially chosen six relations and the accuracy levels for $k@$, $k@10$ and $k@100$ are recorded. The subject labels are injected into the templates, the object labels are masked, and the result is recorded as 1 if the token for the object label is among the top k predicted tokens, otherwise it is labeled as 0. The accuracy is calculated in three ways,

verb-morphology	Template
<i>active</i>	[O] owns [S]. [S] plays [O].
<i>passive</i>	[S] is owned by [O]. [O] is played by [S].
<i>nominalized</i>	[S] is [O]’s property. [O] is [S]’s instrument.

Table 51: Templates for P127, ‘owned by’ (N:1) and P1303, ‘instrument’ (M:N)

first we calculate the total average accuracy over all triplets for all relations, second we calculate the average accuracy of all triplets belonging to the same relation cardinality, and finally we calculate the average performance for each relation individually.

4.3.3 Results

This section describes the results of the experiment for the verb typology, active, passive and nominalization, henceforth referred to as APN. We will not discuss these for all masked language models, but will focus this investigation on the cased versions of the BERT models. These results will clearly show that the models under investigation fact retrieval performance clearly depends on the morphological state of the verb in the sentence. First, we undertake a model comparison in which all models will be compared to one another regarding their accuracy performance on the T-REx data source reduced for APN for all three morphological types of the verb. We analyze the structure of these results, find explanations for them, and attempt to identify other interesting patterns. After this global model comparison, we analyze the results for different metrics before comparing the results for the individual relations. Finally, we will analyze the score values of these models.

4.3.3.1 Model Comparison

This model comparison corroborates the main conclusion of the sentence typology analysis, namely that syntactic structure matters for fact retrieval tasks with LLMs. For all the models considered, the change in verb morphology has an impact on the average accuracy results. Although, in contrast to the results in the sentence typology analysis, models do not have a clear favorite across metrics as the simple sentence is.

Table 52 shows the average k@1 accuracy results for all BERT-based models for the APN reduced T-REx data source. We can immediately see the same performance chiasm we observed for the sentence typology, between the cased models and the uncased models. Therefore, we will once again choose to focus the investigation on the cased version of the BERT-based models. The second immanent observation is the fact that the split in two factions, the uncased and the cases factions, reveals two favorites among the verb morphologies. The passive verb morphology is the clear favorite of the cased models,

whereas the active is the favorite of the uncased models.

Model	Active	Passive	Nominalized
BERT-large-cased	17.966	29.197	19.929
BERT-base-cased	5.949	28.019	13.45
BERT-base-multilingual-cased	9.523	23.267	22.05
BERT-large-uncased	4.025	0	1.1
BERT-base-uncased	1.1	0	0.314
BERT-base-multilingual-uncased	1.571	0.02	0.53

Table 52: APN Model Comparison complete k@1 accuracy

Furthermore, the differences between the models and the verb morphology types are much more pronounced than for the sentence typology. For example, BERT-base-large has much higher k@1 accuracy results for the active and the nominalized types than its smaller counterparts. Both smaller models BERT-base and BERT-base-multilingual have higher k@1 accuracy results for nominalized morphology type than for the active morphology types. This fact is very surprising from the vantage point of human intuition about language complexity. Globally speaking, all cased models clearly prefer passive and then nominalized prompts to active prompts, according to these results.

Table 53 shows the results for the same models, but for the k@10 accuracy. Interestingly and in contrast to the sentence typology, the results change completely when moving from the k@1 to the k@10 accuracy because in the k@10 results the active morphology type is clearly favored by most models, five out of six precisely.

Model	Active	Passive	Nominalized
BERT-large-cased	49.676	37.62	36.776
BERT-base-cased	48.714	36.521	30.022
BERT-base-multilingual-cased	31.278	26.762	33.399
BERT-large-uncased	29.923	0.511	14.903
BERT-base-uncased	11.153	0.314	10.603
BERT-base-multilingual-uncased	12.252	0.334	9.758

Table 53: APN Model Comparison complete k@10 accuracy

An interesting outlier can be identified when examining the active morphology result for BERT-large-uncased because it is the highest result an uncased version of a model has reached thus far, and it even outperforms the passive result for the cased version of BERT-base-multilingual.

4.3.3.2 Metric Comparison

Once again, we focus on the cased versions of the PLMs in the further investigation for the same and already thoroughly discussed reasons as before. In this metric comparison, we analyze the accuracy results for every one of the three cased models. These results

will again clearly demonstrate that the verb morphology of a sentence matters greatly to the results, but in a different way than they do for the sentence typology because we are not able to identify a clear favorite for each of the models.

4.3.3.2.1 BERT-Large-Cased

Table 54 shows the results for the different metrics for BERT-large-cased for the APN verb morphology. These results demonstrate a very diverse picture of the accuracy results because no clear trends emerge from the data. One pattern visible for all accuracy metrics is that there are always two results in close range of each other, and one outlier. The outliers occupy the top scores for k@1 and k@10, but the lowest rank for k@100. The active result gains the most percentage points, going from k@1 to k@10, almost tripling the percentage. Whereas, the nominalized sentences almost double between the k@10 and the k@100 results, which is the highest gain switching between any two metrics. Curiously the passive sentences perform very well, best of the three morphology types, for the k@1 accuracy but gain very little percentage points when more of the top tokens are considered. The pairs with comparable average accuracy levels are active and nominalized for k@1, passive and nominalized for k@10 and active and nominalized again for k@100.

Metric	Active	Passive	Nominalized
k@1 accuracy	17.966	29.197	19.929
k@10 accuracy	49.676	37.62	36.776
k@100 accuracy	79.874	43.471	74.75

Table 54: APN Metric Comparison BERT-large-cased

These results strongly indicate that syntax structure based on the APN verb morphology has a great impact on BERT-large-cased’s fact retrieval performance. Since these results are exceedingly divers across the metrics, the score comparison is of major interest for the APN analysis.

4.3.3.2.2 BERT-Base-Cased

Table 55 shows the results for the different metrics for BERT-base-cased for the APN verb morphology, and it also shows a slightly different pattern compared to the larger BERT-based version. The morphology types in the top positions are in the same place, passive for k@1 and active for k@10 and k100, only with slightly lower accuracy values.

Metric	Active	Passive	Nominalized
k@1 accuracy	5.949	28.019	13.45
k@10 accuracy	48.714	36.521	30.022
k@100 accuracy	76.674	41.488	60.475

Table 55: APN Metric Comparison BERT-base-cased

But the pairs identified for BERT-large-cased do not appear in Table 55. Instead, there exist significant differences in the accuracy levels between the second and third positions. More than 6 percentage points for k@1 and k@10 and almost 20 percentage points for k@100. Table 56 shows the ranking of the verb morphology types based on the average accuracy results for each metric, and we observe that the rankings do not allow for the inference of a stable global ranking of morphology types because the rankings are all different.

Metric	First	Second	Third
k@1 accuracy	passive	nominalized	active
k@10 accuracy	active	passive	nominalized
k@100 accuracy	active	nominalized	passive

Table 56: APN Metric Comparison Accuracy Ranking BERT-base-cased

Once again, we show that syntax structure has an effect on fact retrieval performance on PLMs. Because these results are even more diverse than for the larger BERT-based model, the score comparison is of major interest for the APN analysis.

4.3.3.2.3 BERT-Base-Multilingual-Cased

Table 57 shows the results for the different metrics for BERT-base-multilingual-cased for the APN verb morphology. Table 57 paints an even more diverse picture than its two BERT-based predecessors. For BERT-base-multilingual-cased every one of the three different APN morphology types is in the top spot for one of the accuracy metrics. Thus even less patterns emerge in these results,

Metric	Active	Passive	Nominalized
k@1 accuracy	9.523	23.267	22.05
k@10 accuracy	31.278	26.762	33.399
k@100 accuracy	62.733	33.261	51.463

Table 57: APN Metric Comparison BERT-base-multilingual-cased

The multilingual BERT version further reinforces the indication that syntax structure matters to PLMs when confronted with fact retrieval tasks because there are again large discrepancies in average accuracy results for every metric for all three APN verb morphology types. Concluding the metric comparison entirely, we can state this conclusion across all models investigated here.

4.3.3.3 Relation Comparison

In this paragraph, we take a much closer look at the structure of the average accuracy results by analyzing these results on the level of the individual relations. Because only few of the relations are suitable for the APN verb morphology, we will not differentiate

the results for the different cardinalities. Once more, we will focus on the top performing models, the three cased versions of BERT identified in model comparison and analyzed in the metric comparison. Furthermore, these results once again illustrate that these differences between the fact retrieval performances can not easily be explained by trivial factors.

4.3.3.3.1 BERT-Large-Cased

Table 58 shows the k@1 accuracy results for the APN verb morphology types for the individual relations. We see a very diverse set of results in this table because every APN verb morphology type occupies the top spot for at least a single relation. Furthermore, the results vary drastically between relations and morphology types.

Relation	Active	Passive	Nominalized
Total	17.966	29.197	19.929
P127	9.025	34.643	30.713
P136	1.289	0	3.115
P176	60.285	87.373	66.191
P178	21.284	66.047	21.115
P413	4.097	0	0
P1303	8.851	0	0

Table 58: APN Relation Comparison BERT-large-cased k@1 accuracy

Continuing, Table 59 shows the k@10 accuracy results for the APN verb morphology types for the individual relations and the picture slightly changes as discussed in the metric comparison.

Relation	Active	Passive	Nominalized
Total	49.676	37.62	36.776
P127	45.997	64.192	53.275
P136	22.127	0	43.072
P176	90.53	95.723	81.772
P178	67.23	90.203	39.696
P413	37.08	0	0.84
P1303	38.778	0.105	6.322

Table 59: APN Relation Comparison BERT-large-cased complete k@10 accuracy

Looking at the Tables 58 and 59 in combination, one finds that the passive morphology type exhibits hit or miss style results on a relational level. Either the results are at the top results reached in the experiment, or the results are at 0 or very close to that. Analyzing this phenomenon further, we discover that it is fully coincident with the fact that relations with good passive accuracy results are exactly those for which the prompt template for the passive morphology type is equal to the prompt template for the simple sentence type from the sentence typology. This is interesting in so far as every passive prompt is

also a simple prompt.

4.3.3.3.2 BERT-Base-Cased

Table 60 shows the average k@1 accuracy results for the APN verb morphology types for the individual relations. We again observe a very diverse set of results in this table because every APN verb morphology type occupies the top spot for at least a single relation. Once again, the results vary drastically between relations and morphology types.

Relation	Active	Passive	Nominalized
Total	5.949	28.019	13.45
P127	0.437	34.789	32.46
P136	0.644	0	4.404
P176	23.32	85.642	42.159
P178	1.014	58.615	1.182
P413	0.525	0	0
P1303	5.69	0	0

Table 60: APN Relation Comparison BERT-base-cased k@1 accuracy

Taking a combined look at Table 60 and Table 61 we recognize that the results resemble the structure and patterns of the larger BERT-based model and there are no additional such to identify.

Relation	Active	Passive	Nominalized
Total	48.714	36.521	30.022
P127	28.53	61.863	49.636
P136	44.146	0	45.328
P176	87.984	94.297	70.672
P178	51.858	85.98	11.655
P413	36.975	0	0
P1303	36.986	0	0.316

Table 61: APN Relation Comparison BERT-base-cased complete k@10 accuracy

4.3.3.3.3 BERT-Base-Multilingual-Cased

Table 62 and Table 63 show the average k@1 and k@10 accuracy results for the APN verb morphology types for the individual relations. These results, too, resemble the structure and patterns of the other BERT-based models, and thus there are no additional interesting patterns to discuss.

4.3.3.4 Score Comparison

Relation	Active	Passive	Nominalized
Total	9.523	23.267	22.05
P127	6.696	17.467	16.157
P136	0.644	0	21.482
P176	37.678	83.401	72.709
P178	6.081	41.554	16.554
P413	0	0	0
P1303	2.845	0	0

Table 62: APN Relation Comparison BERT-base-multilingual-cased k@1 accuracy

Relation	Active	Passive	Nominalized
Total	31.278	26.762	33.399
P127	23.29	25.91	24.6
P136	22.342	0	51.343
P176	80.346	90.428	83.096
P178	47.804	49.662	37.162
P413	2.311	0	0.315
P1303	13.804	0.316	1.581

Table 63: APN Relation Comparison BERT-base-multilingual-cased k@10 accuracy

For a second time, we will move on from the analysis of the accuracy metrics and thus from results based in rankings to the source of these rankings, the token scores themselves. Of these scores, we are, once more, most interested in the scores the model assigns to the correct token for each triple. In this paragraph we will show and analyze the correct token scores and compare them for the different APN morphology types and cardinalities and construct a probability distribution for each of those form and compare these using the KL-Divergence.

4.3.3.4.1 BERT-Large-Cased

Table 64 shows the average correct token scores for BERT-large-cased for all relations in total, grouped by the different relational cardinalities and for the individual relations. In contrast to the different accuracy metrics, this perspective offers a clear favorite APN verb morphology because the passive morphology type has the highest values in total, for all relevant cardinality types and for half the relations. We can once more identify the hit or miss property of the passive verb morphology because it either reaches the highest score or has a correct token probability of 0.

From the correct token scores for the APN verb morphology, we construct a probability distribution for each of the three morphology types by counting the number of scores within 99 consecutive intervals between 0 and 1. Figure 20 shows plots of these three probability distributions.

Since Figure 20 only shows once more that most of the probability mass is located around

Name	Active	Passive	Nominalized
Total	0.019	0.063	0.019
N:1	0.024	0.075	0.026
N:M	0.005	0.036	0.001
P413	0.01	0	0.001
P127	0.003	0.059	0.028
P136	0.006	0	0.014
P176	0.071	0.231	0.062
P178	0.007	0.095	0.002
P1303	0.004	0	0

Table 64: APN Correct Token Scores BERT-large-cased

0, Figure 21 shows a zoomed version of the plot that shows more details about how the models do differ from each other. In the zoomed in plot of the passive morphology type’s probability distribution, we find the same second local maximum near 1 at the end of the distribution. Compared to the passive morphology type’s the active and nominalized morphology types are very similar to each other.

In order to extend this comparison of probability distributions, we once more construct a distribution comparison based on the KL-Divergence. Figure 22 shows a heatmap constructed from the different values for the KL-Divergence of the probability distributions for two respective morphology types. Figure 22 clearly shows that the highest discrepancy exists between the passive morphology types and the other two, whereas the non-passive sentence types are much better substitutes for each other.

Finally, we can state that syntax structure in the form of verb morphology has a sizable impact on the BERT-larges’ correct token scores and thus for fact retrieval performance in general. The passive verb morphology’s probability distribution emerges as the unique one of the three candidates.

4.3.3.4.2 BERT-Base-Cased

Moving on to the next model in our analysis of the correct token scores, Table 65 shows the average correct token scores for BERT-base-cased for all relations in total, grouped by the different relational cardinalities and for the individual relations. In congruence with the previously analyzed model, the passive morphology type has the highest correct token probability values in total, for all relevant cardinality types and for half the relations. We can also identify the hit or miss property of the passive verb morphology, again because it either reaches the highest score or has a correct token probability of 0.

From the correct token scores for the APN verb morphology, we also construct a probability distribution for each of the three morphology types by the same procedure as before. Figure 23 shows plots of these three probability distributions.

Since Figure 23 again only shows very little, namely that most of the probability mass is located around 0, Figure 24 shows a zoomed version of the plot that shows more details

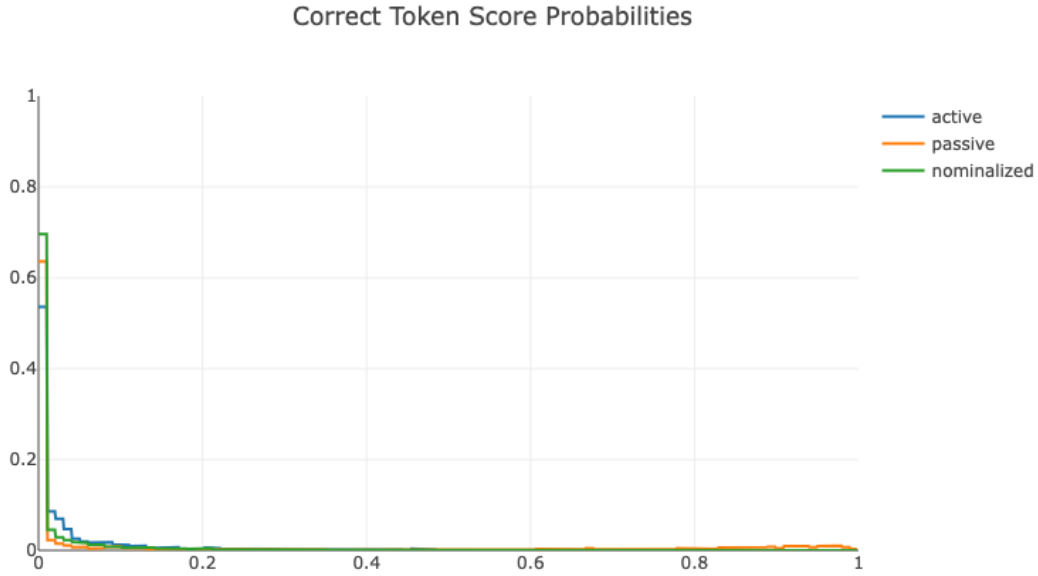


Figure 20: APN Score Probability Distribution BERT-large-cased

about how the models differ from each other. In the zoomed in plot of the passive morphology type’s probability distribution, we find the same second local maximum near 1 at the end of the distribution. Compared to the passive morphology type’s the active and nominalized morphology types are very similar to each other. Curiously, this plot is identical to the plot based on the large version of BERT.

We again construct a distribution comparison based on the KL-Divergence. Figure 28 shows a heatmap constructed from the different values for the KL-Divergence of the probability distributions for two respective morphology types. Figure 28 clearly shows that the highest discrepancy exists between the passive morphology types and the other two, whereas the non-passive sentence types are much better substitutes for each other. These results are identical to results seen for the large version of BERT.

4.3.3.4.3 BERT-Base-Multilingual-Cased

Moving on to the final model in our analysis of the correct token scores, Table 66 shows the average correct token scores for BERT-base-multilingual-cased for all relations in total, grouped by the different relational cardinalities and for the individual relations. In congruence with both previously analyzed models, the passive morphology type has the highest correct token probability values in total, for all relevant cardinality types and for half of the relations. We can also identify the hit or miss property of the passive verb morphology for a third time.

For a final time, we construct a probability distribution for each of the three morphology

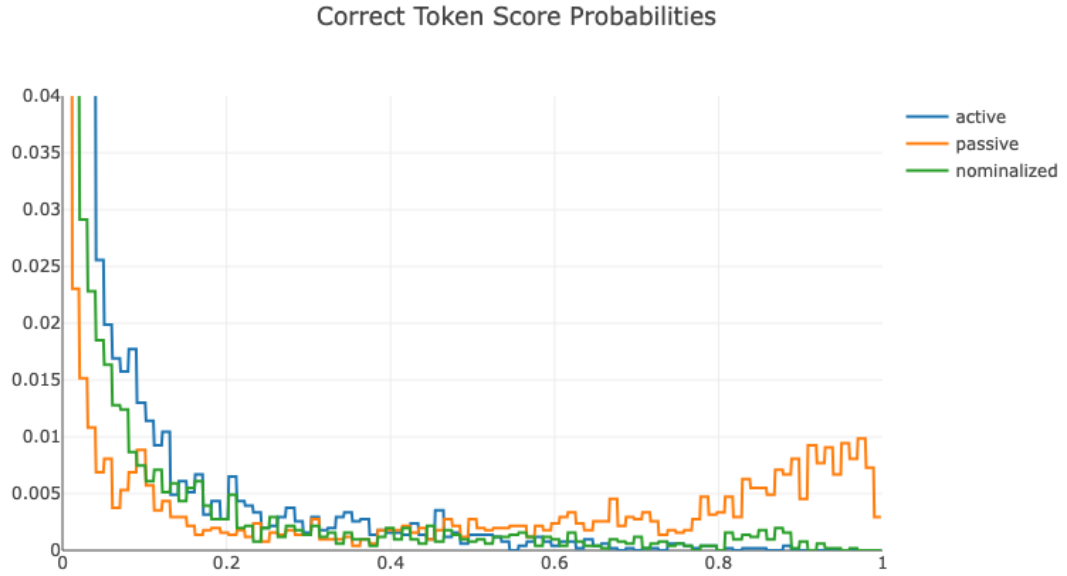


Figure 21: APN Score Probability Distribution Zoomed BERT-large-cased

types from the correct token scores for the APN verb morphology, by the same procedure as before. Figure 26 shows plots of these three probability distributions and Figure 27 shows the zoomed version of these plots

Additionally, we end this analysis with a distribution comparison based on the KL-Divergence. Figure 28 shows a heatmap constructed from the different values for the KL-Divergence of the probability distributions for two respective morphology types. Figure 28 clearly shows that the highest discrepancy exists between the passive morphology types and the other two, whereas the non-passive sentence types are much better substitutes for each other. These results are identical to results seen for the large version of BERT.

4.3.3.5 Embedding Comparison

So far we have compared different models according to accuracy, analyzed different accuracy values for the top performing models, analyzed the results for different segments of the data set in detail and analyzed the scores assigned by the models to the correct tokens for the three different APN morphology types. Finally, we continue to the analysis of models embeddings. The embeddings are the final representation of the input as processed by the model. They are the final hidden states and since they are used further as input to downstream tasks in fine-tuning, they are often called the features of a given input. Thus, these features represent the model’s final interpretation. For this reason, we analyze and compare these features for the three different morphology types for the

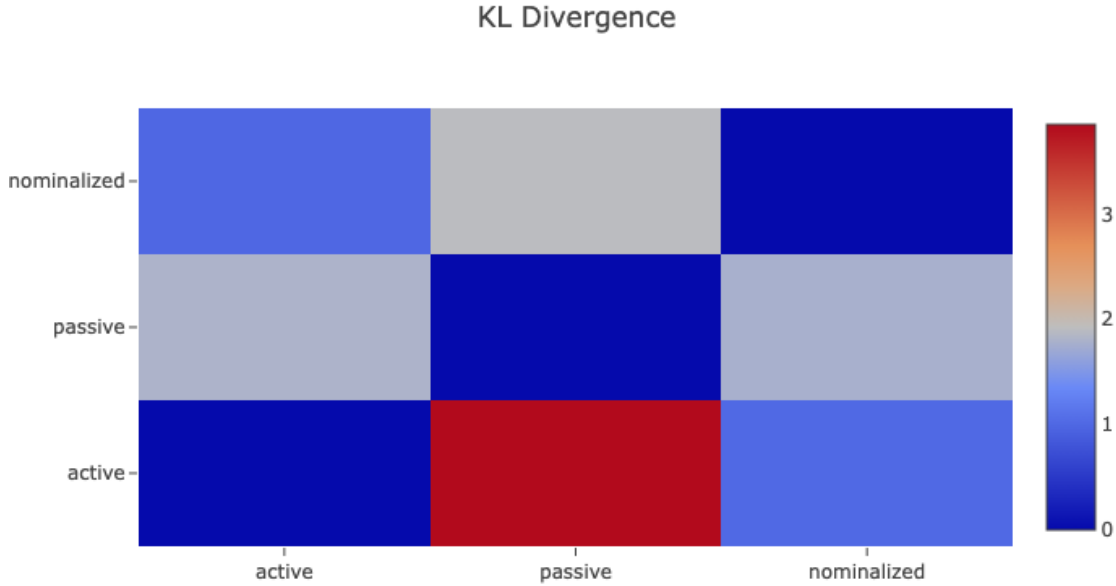


Figure 22: APN KL-Divergence BERT-large-cased

chosen PLMs. In order to be able to compare these features, we will use the metric cosine similarity to again ask how similar the embeddings for the different verb morphology types are to each other by calculating the cosine similarity between the average embeddings per morphology type.

Figure 29 shows a heat map constructed from the cosine similarities between the embeddings of the different morphology types. The cosine similarity measure is a measure that projects the values into the interval $[-1, 1]$. Thus, the fact the similarity values span only from 0.5 to 1 is already an indication that the embeddings themselves are not completely dissimilar. This is interesting because so far we did not know anything about the inner representation of the morphology types, except for the property if the model is able to predict the correct token. But generally, it would have been possible that the model reached two correct predictions based on two completely dissimilar sets of features. But the model seems to understand in some way that the sentences, which are completely equal in semantics and only differ in syntax, are themselves quite similar. This conclusion is analogous to the conclusion made about the sentence typology. Figure 30 and 31 show the same embedding comparison for the other two bio-based model and substantiate this conclusion.

5 Conclusion

The central question of this work is, how syntax structure impacts fact retrieval performance in Pre-trained Language Models. More concretely, we ask what differences occur

Name	Active	Passive	Nominalized
Total	0.019	0.063	0.019
N:1	0.024	0.075	0.026
N:M	0.005	0.036	0.001
P413	0.01	0	0.001
P127	0.003	0.059	0.028
P136	0.006	0	0.014
P176	0.071	0.231	0.062
P178	0.007	0.095	0.002
P1303	0.004	0	0

Table 65: APN Correct Token Scores BERT-base-cased

Name	Active	Passive	Nominalized
Total	0.019	0.063	0.019
N:1	0.024	0.075	0.026
N:M	0.005	0.036	0.001
P413	0.01	0	0.001
P127	0.003	0.059	0.028
P136	0.006	0	0.014
P176	0.071	0.231	0.062
P178	0.007	0.095	0.002
P1303	0.004	0	0

Table 66: VerbTypology Correct Token Scores BERT-base-multilingual-cased

and why do differences occur when PLMs are confronted with syntactically different but semantically equivalent prompts. In order to answer this question, we choose the sentence typology and the APN verb morphology as linguistic frameworks to construct semantically equivalent but syntactically different prompt templates for the T-REx data source and compared fact retrieval performance for an array of different models, mostly BERT-based.

In conclusion, syntax has an influence on the performance of PLMs. Most prominent language models, when confronted with syntactically different but semantically equivalent prompts, answer with very different token probabilities. Simple and not nominalized sentences reliably outperform other syntactical prompt variants. For the sentence typology, the results are very clear. Every vantage point on the results offered the same conclusion, the simple sentences achieve the highest fact retrieval performance results for every metric and for every investigated slice of the data, except for some single relations. This is explained by the comparison of the different probability distribution of the correct token scores, because the probability mass of the simple sentence’s distribution is shifted to the right, towards the value 1, compared to the other distributions. A similar phenomenon can be observed for the APN verb morphology. The models under investigation all preferred the passive verb morphology to the active, and especially to the nominalized verb morphology. But the results are not as clear as for the sentence typology, for two reasons.

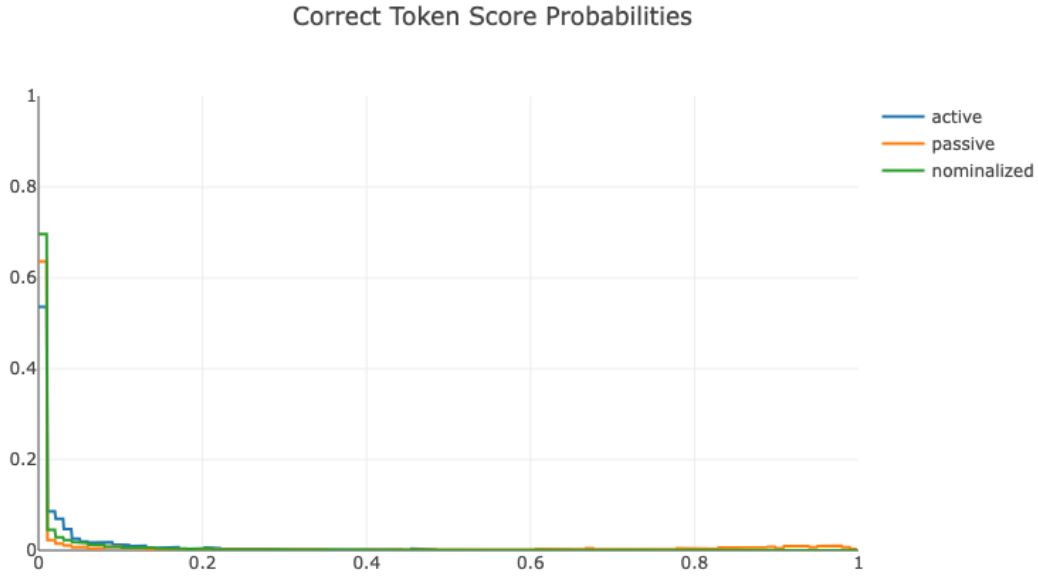


Figure 23: APN Score Probability Distribution BERT-large-cased

Firstly, the data set had to be greatly reduced in order to conduct the analysis. Secondly, the passive verb morphology has a superior fact retrieval performance than the other types in total, but not for all relations. In cases where the passive verb morphology is not the best verb morphology, the active verb morphology reaches the top spot in the ranking and the passive verb morphology’s accuracy levels drop to zero, demonstrating a hit or miss property for the passive verb morphology based prompts. As expected, the largest of the BERT-based models outperforms all other models on all syntax categories under investigation.

The sentence typology offers a way to construct syntactically different but semantically equivalent prompts in order to assess the relationship between syntax and fact retrieval performance of PLMs. The experiments conducted in this work can only serve an explorative first step in analyzing the relationship between syntax and PLMs for several reasons. First, there are multifarious ways to construct syntactically different but semantically equivalent prompts, and if one starts to consider only relevant semantic information even more possibilities emerge. Furthermore, we only use a single representative for each of the four different sentence types in each relational comparison and thus completely ignore the possibility of huge differences in performance between different prompts that belong to the same sentence type. This phenomenon can be observed in the APN verb morphology results because the active and passive morphology types’ templates both always belong to the category of a simple sentence according to the sentence typology. In the future, it would be intriguing to include more templates for each of the four sentence types. A possible avenue of investigation could be to repeat the type comparison by averaging the results for each of the sentence types.

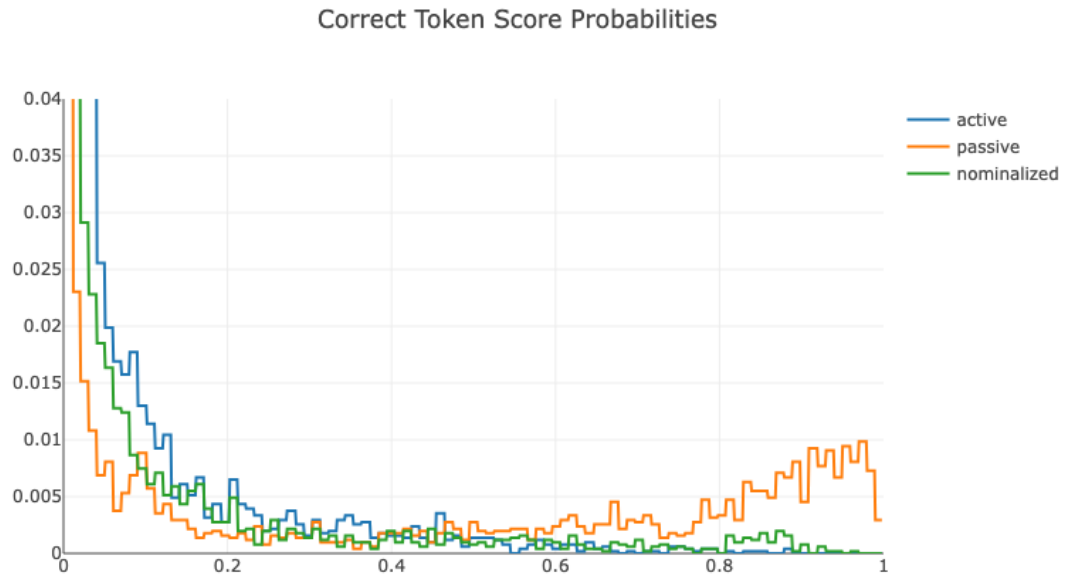


Figure 24: APN Score Probability Distribution Zoomed BERT-base-cased

Additionally, although many researchers have also chosen the T-REx data source as the data source of choice in fact retrieval experiments, it seems that its composition hinders the explanatory power of this work’s results. Since it comprises mostly named and not abstract entities, the uncased versions of many models are excluded in this analysis because of their low accuracy levels in the experiments. Regrettably, a focus on just the abstract entities would have left us with not enough of a data size to meaningful conclusions about their relationship between syntax and performance. It would be very interesting to perform this analysis on a large data source comprised of triples of only abstract entities and comparing the results. This would have the added benefit that models that are largely excluded in this investigation could be included into the comparison, especially SenseBERT and Electra, for which cased version are not publicly available at the time of this writing.

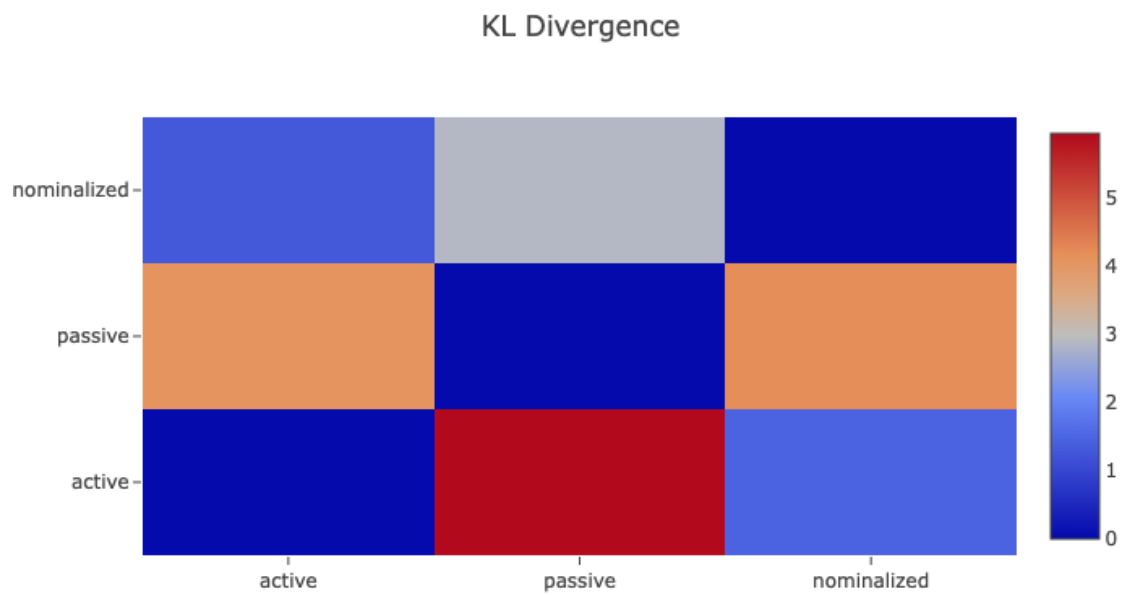


Figure 25: APN KL-Divergence BERT-base-cased

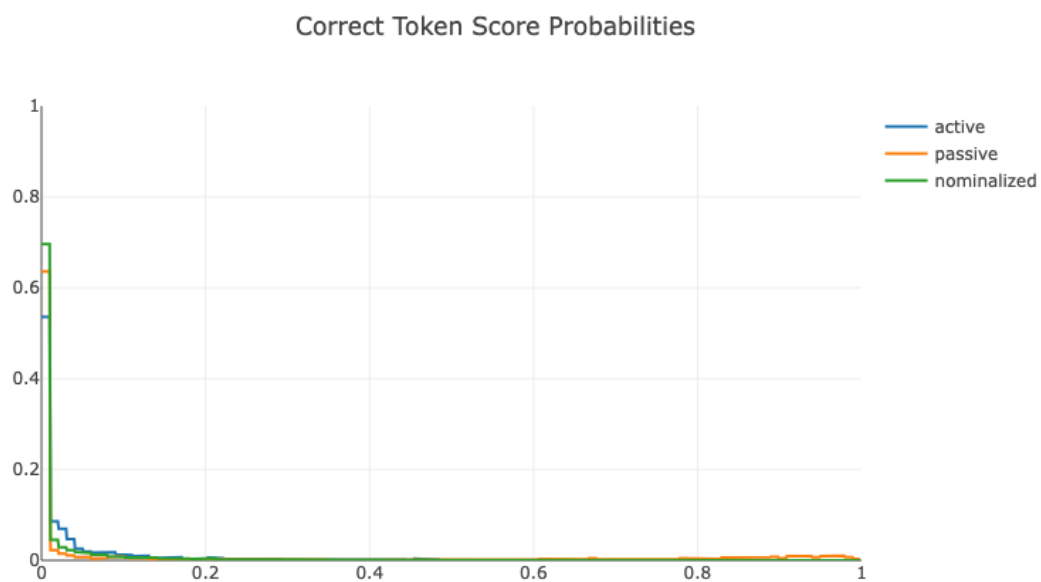


Figure 26: APN Score Probability Distribution BERT-base-multilingual-cased

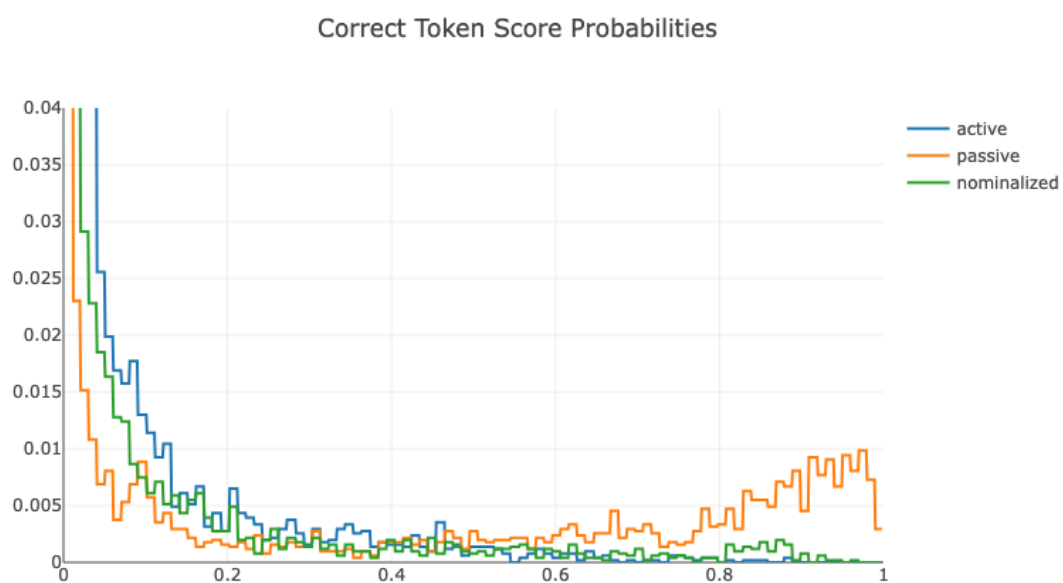


Figure 27: APN Score Probability Distribution Zoomed BERT-base-multilingual-cased

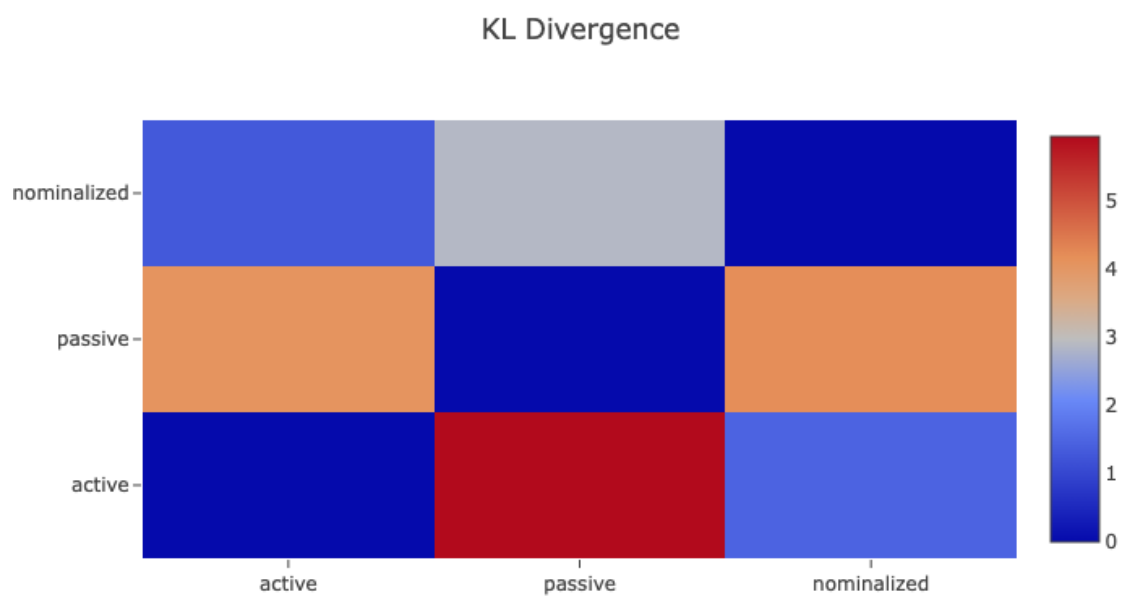


Figure 28: APN KL-Divergence BERT-base-multilingual-cased

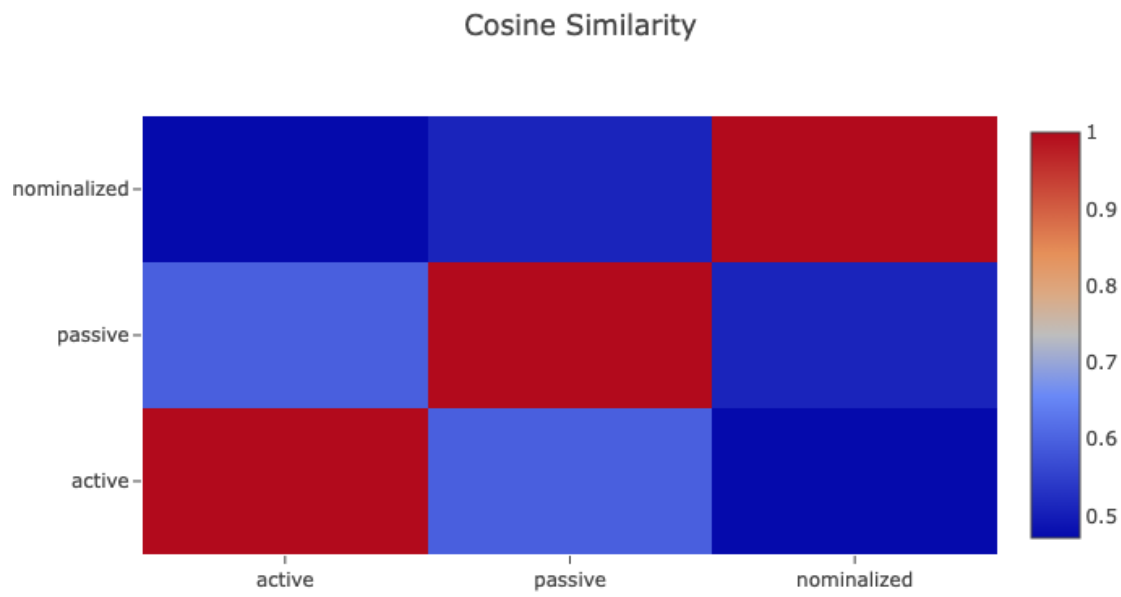


Figure 29: APN Cosine Similarity Heat Map for bert-large-cased

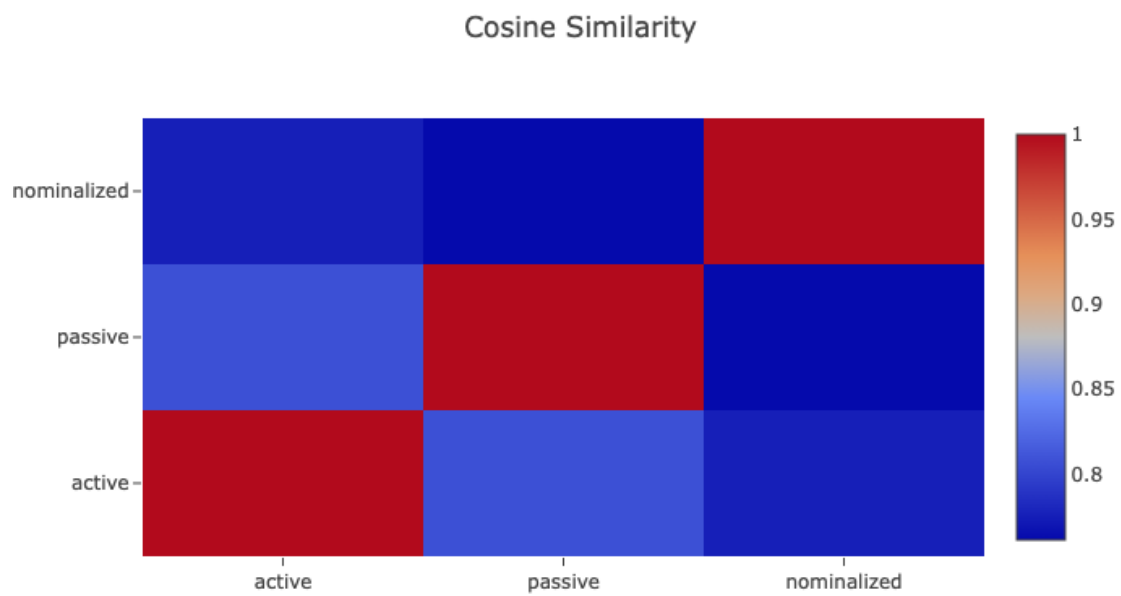


Figure 30: APN Cosine Similarity Heat Map for bert-base-cased

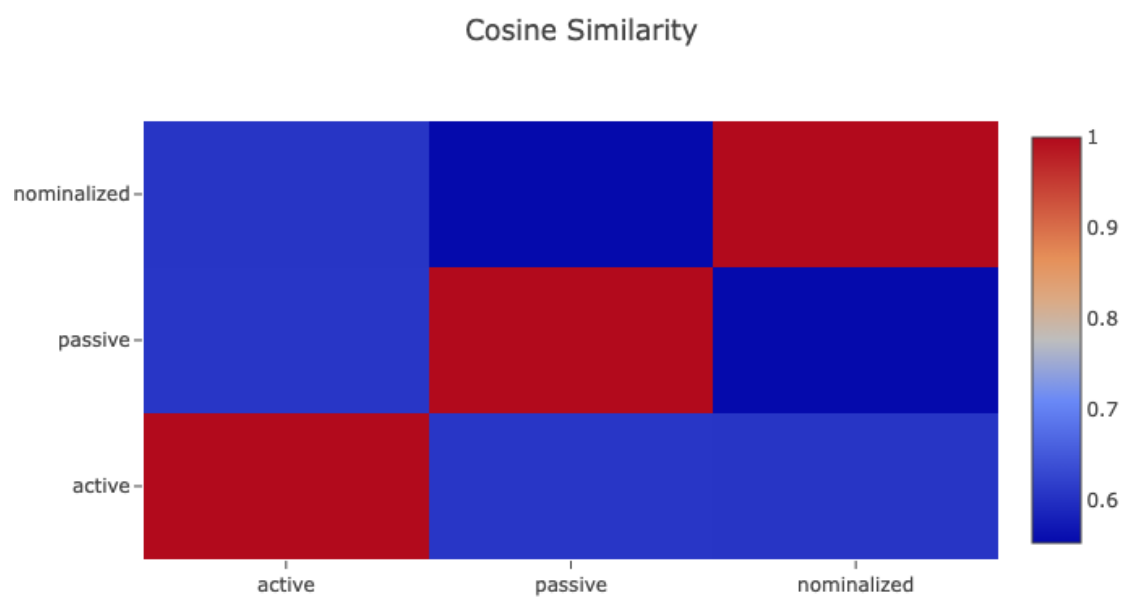


Figure 31: APN Cosine Similarity Heat Map for bert-base-multilingual-cased

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6 Appendix

6.1 Templates

Sentence Typology

- P159: simple: The headquarter of [S] is in [O].
- P159: compound: [S] is located in [O] and that is where they have their headquarters
- P159: complex: [S] is located in [O] because they have their headquarters there.
- P159: compound-complex: [S] has a location in [O] and this location is important because it is their headquarters.
- P37: simple: The official language of [S] is [O].

- P37: compound: [O] is used in [S] and it is also its official language.
- P37: complex: [O] is used in [S] which is also its official language.
- P37: compound-complex: [O] is a language and it is used in [S] which is also its official one.
- P1412: simple: [S] used to communicate in [O].
- P1412: compound: [O] is a language and [S] used to communicate in it.
- P1412: complex: [O] is the language that [S] used to communicate in.
- P1412: compound-complex: [O] is a language which many people used to communicate in and [S] did exactly that.
- P138: simple: [S] is named after [O].
- P138: compound: Many things were named after [O] and [S] is an example of that.
- P138: complex: Many things were named after [O] which [S] is an example of.
- P138: compound-complex: Many things were named after [O] and [S], among other things, is an example of that.
- P495: simple: [S] was created in [O].
- P495: compound: [O] is a country and [S] was created in it.
- P495: complex: [O] is the country that [S] was created in.
- P495: compound-complex: [O] is a country and it is the one that [S] was created in.
- P17: simple: [S] is located in [O].
- P17: compound: [O] is a country and [S] is located there.
- P17: complex: [S] is located in [O], which is a country.
- P17: compound-complex: [O] is a country and it is comprised of many locations, for example [S].
- P178: simple: [S] was created by [O] .
- P178: compound: [O] has created something and it is called [S].
- P178: complex: [O] has created something that is called [S].
- P178: compound-complex: [O] creates things and there is one example that is called [S].
- P103: simple: The native language of [S] is [O].
- P103: compound: [S] speaks [O] and it is also his native language.
- P103: complex: [S] has a native language which is [O].

- P103: compound-complex: [S] speaks [O] and he speaks it well because it is his native language.
- P20: simple: [S] died in [O].
- P20: compound: [O] is the name of a place and [S] died there.
- P20: complex: [O] is the name of the place, where [S] died.
- P20: compound-complex: [O] is the name of the place, where some people died, for example [S].
- P36: simple: The capital of [S] is [O] .
- P36: compound: [S] is a country and it's capital is [O].
- P36: complex: [S] is the country, who's capital is [O].
- P36: compound-complex: [O] is a city and it is the city that is the capital of [S].
- P407: simple: [S] was written in [O].
- P407: compound: [O] is a language and it was used to write [S].
- P407: complex: [O] is a language, which was used to write [S].
- P407: compound-complex: [O] is a language in which many works were written in and it was also used to write [S].
- P190: simple: [S] and [O] are twin cities.
- P190: compound: [S] and [O] are both cities and they are also twin cities.
- P190: complex: [S] and [O] are twinned administrative bodies, which means they are twin cities.
- P190: compound-complex: [S] and [O] are both administrative bodies and they are twinned, which means they are twin cities.
- P276: simple: [S] is located in [O].
- P276: compound: [S] is a place and it is located in [O].
- P276: complex: [S] is a place which is located in [O].
- P276: compound-complex: [S] is a place that is well known and it is located in [O].
- P108: simple: [S] works for [O].
- P108: compound: [O] is an employer and as such it employs [S].
- P108: complex: [S] has an employer, which is [O].
- P108: compound-complex: [O] employs many people and [S] is an example because he works for them.

- P31: simple: [S] is a [O].
- P31: compound: [S] is an instance of something and it is one of [O].
- P31: complex: [S] is an instance of something, which is [O].
- P31: compound-complex: [O] has many instances of it and one is called [S], which is one.
- P364: simple: The original language of [S] is [O].
- P364: compound: [S] can be watched in its original language and it is [O].
- P364: complex: [S] can be watched in its original language which is [O].
- P364: compound-complex: The production was called [S] and it can be watched in its original language, which is [O].
- P27: simple: [S] is [O] citizen.
- P27: compound: [O] is a country and [S] is a citizen of it.
- P27: complex: [O] is the country of which [S] is a citizen.
- P27: compound-complex: [O] is a country, with many citizens, for example [S] is one of them.
- P30: simple: [S] is located in [O].
- P30: compound: [O] is a continent and [S] is located there.
- P30: complex: [O] is the continent in which [S] is located.
- P30: compound-complex: [O] is a continent and it is the one in which [S] is located.
- P47: simple: [S] shares border with [O].
- P47: compound: [S] is a country and it shares a border with [O].
- P47: complex: [S] is one of the countries that shares a border with [O].
- P47: compound-complex: [O] and [S] are countries and have a border that they share.
- P1376: simple: [S] is the capital of [O].
- P1376: compound: [S] is a city it and it is the capital of [O].
- P1376: complex: [S] is the city that is the capital of [O].
- P1376: compound-complex: [S] is a city and it is the city that is the capital of [O].
- P106: simple: [S] is a [O] by profession.
- P106: compound: [S] was known in his field of work and his particular field was [O].

- P106: complex: [S] has a profession, which is [O].
- P106: compound-complex: [S] was known in his discipline for the work he had done in his field and the subject of his work was [O].
- P413: simple: [S] plays in [O] position.
- P413: compound: [O] is a position and [S] plays in it.
- P413: complex: [O] is the position that [S] plays in.
- P413: compound-complex: [O] is one position out of many and it is the one that [S] plays in.
- P937: simple: [S] used to work in [O].
- P937: compound: [O] is a location and [S] used to work there.
- P937: complex: [O] is the location where [S] used to work in.
- P937: compound-complex: [O] is a location and it is the one where [S] used to work in.
- P449: simple: [S] was originally aired on [O].
- P449: compound: [O] is a broadcaster and [S] aired on it.
- P449: complex: [O] is the broadcaster, that [S] aired on.
- P449: compound-complex: [O] is one broadcaster out of many and it is the one, that [S] aired on.
- P131: simple: [S] is located in [O].
- P131: compound: [O] is a location and [S] is located in it.
- P131: complex: [O] is the location in which [S] is located.
- P131: compound-complex: [O] is a location and an example of something located in it is [S].
- P1303: simple: [S] plays [O].
- P1303: compound: [O] is an instrument and [S] plays it.
- P1303: complex: [O] is an instrument, which is played by [S].
- P1303: compound-complex: [O] is an instrument and it is associated with [S] because he plays it.
- P127: simple: [S] is owned by [O].
- P127: compound: [S] has an owner and it' is [O].
- P127: complex: [S] has an owner which is [O].

- P127: compound-complex: [O] owns many things and [S] is an example because it is owned by them.
- P279: simple: [S] is a subclass of [O].
- P279: compound: [O] has subclasses and [S] is an example of that.
- P279: complex: [S] is a subclass of another class which is [O].
- P279: compound-complex: [O] is a class and it has subclasses, like [S].
- P740: simple: [S] was founded in [O].
- P740: compound: [O] is a location and [S] was founded there.
- P740: complex: [O] is the location where [S] was founded
- P740: compound-complex: [O] is a location and it is the one where [S] was founded.
- P19: simple: [S] was born in [O].
- P19: compound: [O] is a place and [S] was born there.
- P19: complex: [O] is the place where [S] was born.
- P19: compound-complex: [O] is the name of the place, where some people were born for example [S] was born there.
- P527: simple: [S] consists of [O].
- P527: compound: [S] has parts and [O] is one of them.
- P527: complex: [S] has parts many parts, for example [O].
- P527: compound-complex: [S] has parts and [O], like many others, is one of them.
- P136: simple: [S] plays [O] music.
- P136: compound: [O] is a genre and [S] fits into it.
- P136: complex: [S] is part of [O], which is a popular genre.
- P136: compound-complex: [O] is a genre comprised of many creators and their work, and [S] is an example of such.
- P264: simple: [S] is represented by music label [O].
- P264: compound: [O] is a music label and it represents [S].
- P264: complex: [O] represents [S] because it is a music label.
- P264: compound-complex: [O] is a music label, which means it represents musicians and it represents [S] for example.
- P1001: simple: [S] applies to the jurisdiction of [O].

- P1001: compound: [O] is a jurisdiction and [S] applies to it.
- P1001: complex: [S] is applied to by a certain jurisdiction which is [O].
- P1001: compound-complex: [S] applies to [O] because the later is a jurisdiction and the former applies to it.
- P39: simple: [S] has the position of [O].
- P39: compound: [O] is a position and it is held by [S].
- P39: complex: [O] is a position which is held by [S].
- P39: compound-complex: [O] is a position and it is occupied because it is held by [S].
- P101: simple: [S] works in the field of [O] .
- P101: compound: [S] was known in his field of work and his particular one was [O].
- P101: complex: People in his time knew [O]'s field of work, which was [S].
- P101: compound-complex: [S] was known in his field for the work that he had done and his work was in [O].
- P140: simple: [S] is affiliated with the religion called [O].
- P140: compound: [O] is a religion or a worldview and [S] is affiliated with it.
- P140: complex: [S] is affiliated with [O] which is a religion or a worldview.
- P140: compound-complex: Many people are associated with a religion or a worldview and [S] is affiliated with one of these, which is [O] specifically.
- P361: simple: [S] is part of [O].
- P361: compound: [S] is part of another thing and that thing is [O].
- P361: complex: [S] is part of another thing which is [O].
- P361: compound-complex: [O] has many parts, that make it up and [S] is of them.
- P530: simple: [S] maintains diplomatic relations with [O].
- P530: compound: [S] maintains diplomatic relations with many countries and [O] is one of them.
- P530: complex: [S] is a country that maintains diplomatic relations with [O].
- P530: compound-complex: [S] is a country that maintains diplomatic relations with many others and [O] is one of them.
- P463: simple: [S] is a member of [O].

- P463: compound: [O] is an organisation and [S] is a member of it.
- P463: complex: [O] is one organisation that [S] is a member of.
- P463: compound-complex: [O] is an organisation with many members and [S] is one of them.
- P176: simple: [S] is produced by [O].
- P176: compound: [O] is a producer and [S] is one of their products.
- P176: complex: [S] has a producer which is [O].
- P176: compound-complex: [O] is a producer and [S] is a product that is made by them.

APN Verb Morphology

- P127: active: [O] owns [S].
- P127: passive: [S] is owned by [O].
- P127: nominalized: [S] is [O]'s property.
- P136: active: [S] plays [O] music.
- P136: passive: [O] music is played by [S].
- P136: nominalized: [S] is a [O] musician.
- P176: active: [O] produces [S].
- P176: passive: [S] is produced by [O].
- P176: nominalized: [S] is a [O] product.
- P178: active: [O] created [S].
- P178: passive: [S] was created by [O].
- P178: nominalized: [S] was a [O] creation.
- P413: active: [S] plays in [O] position.
- P413: passive: Position [O] is played by [S].
- P413: nominalized: [S] is the [O] position player.
- P1303: active: [S] plays [O].
- P1303: passive: [O] is played by [S].
- P1303: nominalized: [O] is [S]'s instrument.

6.2 Code

All templates as well as all code that was written to perform the experiments is available online on GitHub. Link: <https://github.com/Thrasolt/ContextualKnowledgeOfLMs>